

LECTURE-8

Photosynthesis



INDRAPRASTHA INSTITUTE *of*
INFORMATION TECHNOLOGY
DELHI

Photosynthesis



10

Photosynthesis

KEY CONCEPTS

- 10.1 Photosynthesis converts light energy to the chemical energy of food
- 10.2 The light reactions convert solar energy to the chemical energy of ATP and NADPH
- 10.3 The Calvin cycle uses the chemical energy of ATP and NADPH to reduce CO_2 to sugar
- 10.4 Alternative mechanisms of carbon fixation have evolved in hot, arid climates

Chloroplast: Capture light energy and convert it to chemical energy.

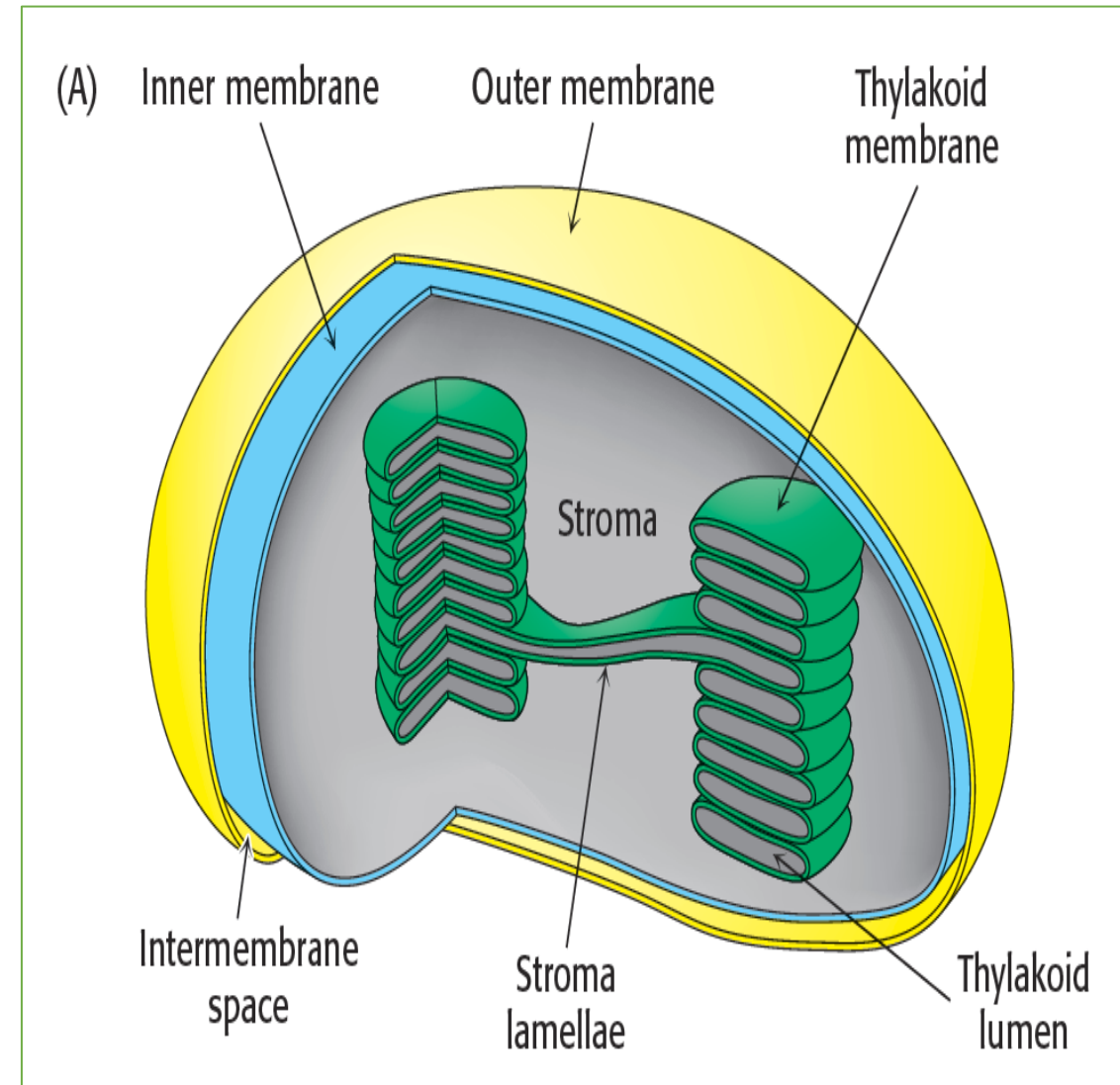


- ❖ Chloroplasts **contain the green pigment chlorophyll** as well as enzymes and other molecules that function in the photosynthetic production of sugar.
 - ❖ *They are 3–6 μm in diameter and are found in leaves and other green organs of plants and algae.*
- ❖ The contents of the chloroplast are separated from the cytosol by an envelope consisting of two membranes separated by a narrow intermembrane space.
- ❖ Inside the innermost membrane is a fluid-filled space, the **stroma**, in which float membranous sacs, the **thylakoids**.
 - ❖ *The stroma contains chloroplast DNA, ribosomes, and enzymes.*
 - ❖ *The thylakoids are flattened sacs that play a critical role in converting light to chemical energy. In some regions, **thylakoids are stacked like poker chips into grana.***

Chloroplast



- ❖ The membranes of the chloroplast divide the chloroplast into **three compartments**: the intermembrane space, the stroma, and the thylakoid space.
- ❖ Like mitochondria, chloroplasts are dynamic structures.
 - ✓ *Their shape is plastic. They are mobile and move around the cell along tracks of the cytoskeleton.*
- ❖ The chloroplast belongs to a **family of plant structures called plastids**.
 - ❖ *Amyloplasts are colorless plastids that store starch in roots and tubers.*
 - ❖ *Chromoplasts store pigments for fruits and flowers.*



Photosynthesis–Light to chemical energy



- ❖ The equation describing the process of photosynthesis is
 - ❖ $6\text{CO}_2 + 12\text{H}_2\text{O} + \text{light energy} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 + 6\text{H}_2\text{O}$ (anabolic process)
- ❖ $\text{C}_6\text{H}_{12}\text{O}_6$ is glucose, although the direct product of photosynthesis is actually a **three-carbon sugar** that can be used to make glucose.
- ❖ *Water appears on both sides of the equation because 12 molecules of water are consumed and 6 molecules are newly formed during photosynthesis.*
- ❖ We can simplify the equation by showing only the net consumption of water:
 - ✓ $6\text{CO}_2 + 6\text{H}_2\text{O} + \text{light energy} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$

Photosynthesis–reverse of cellular respiration



- ❖ Written this way, **the overall chemical change during photosynthesis is the reverse of cellular respiration.**
- ❖ Both of these metabolic processes occur in plant cells. However, chloroplasts do not synthesize sugars by simply reversing the steps of respiration.
- ❖ In its simplest possible form, **$\text{CO}_2 + \text{H}_2\text{O} + \text{light energy} \rightarrow [\text{CH}_2\text{O}] + \text{O}_2$** , where $[\text{CH}_2\text{O}]$ represents the general formula for a carbohydrate.

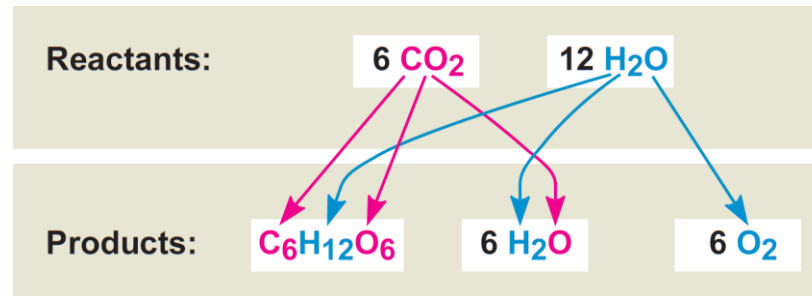
Chloroplasts split water molecules



- ❖ Before the 1930s, *the prevailing hypothesis* was that photosynthesis **split carbon dioxide and then added water to the carbon**:
 - ❖ Step 1: $\text{CO}_2 \rightarrow \text{C} + \text{O}_2$
 - ❖ Step 2: $\text{C} + \text{H}_2\text{O} \rightarrow \text{CH}_2\text{O}$
- ❖ Stanford University's **van Niel challenged this hypothesis**. In the bacteria that he was studying, hydrogen sulfide (H_2S), rather than water, is used in photosynthesis.
- ❖ These bacteria produce yellow globules of sulfur as a waste, rather than oxygen.
- ❖ He proposed this chemical equation for photosynthesis in sulfur bacteria:
 - ❖ $\text{CO}_2 + 2\text{H}_2\text{S} \rightarrow [\text{CH}_2\text{O}] + \text{H}_2\text{O} + 2\text{S}$
- ❖ **He generalized this idea and applied it to plants**, proposing this reaction for their photosynthesis:
 - ❖ $\text{CO}_2 + 2\text{H}_2\text{O} \rightarrow [\text{CH}_2\text{O}] + \text{H}_2\text{O} + \text{O}_2$

Water as a source of electrons from hydrogen atoms

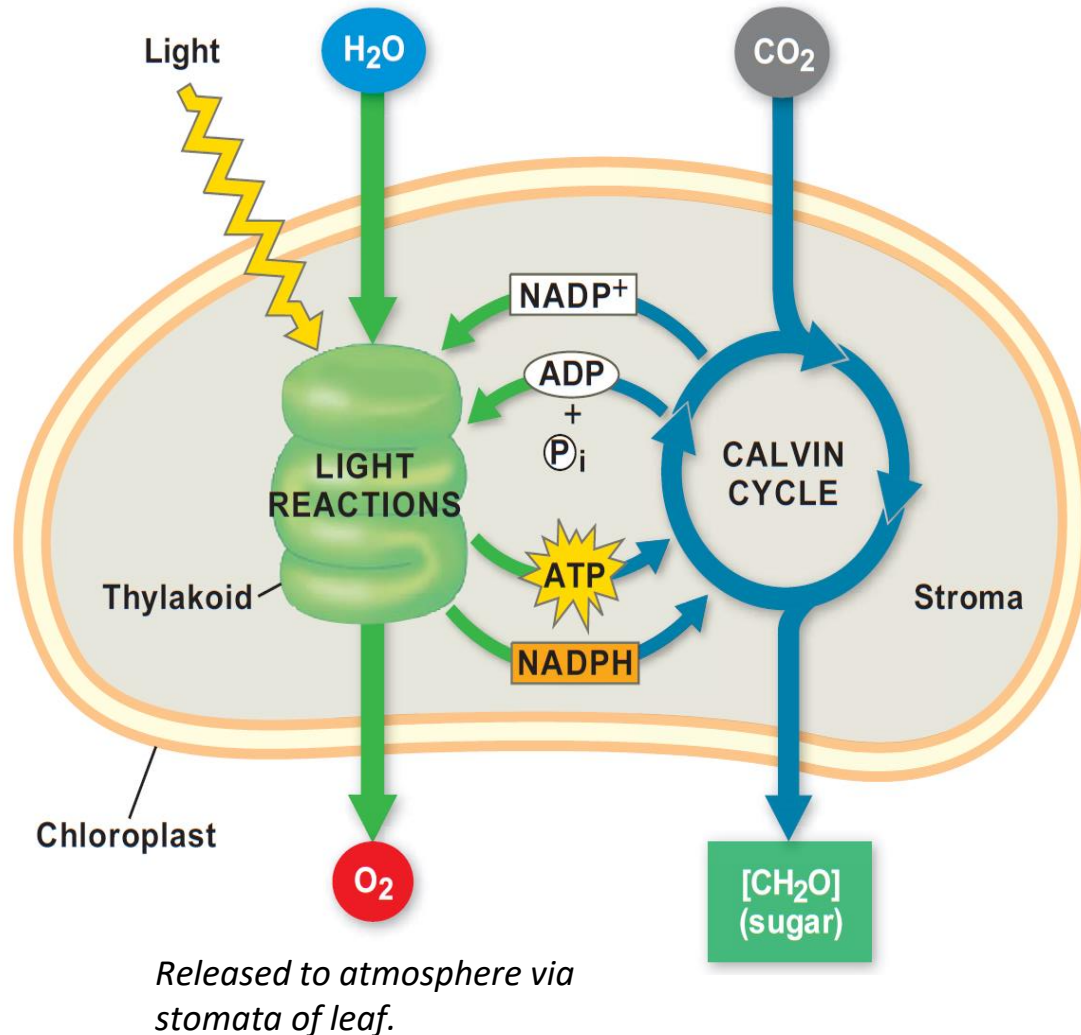
- ❖ Thus, van **Niel hypothesized that plants split water** as a source of electrons from hydrogen atoms, **releasing oxygen as a by-product**.
 - ✓ Sulfur bacteria: $\text{CO}_2 + 2\text{H}_2\text{S} \rightarrow [\text{CH}_2\text{O}] + \text{H}_2\text{O} + 2\text{S}$
 - ✓ Plants: $\text{CO}_2 + 2\text{H}_2\text{O} \rightarrow [\text{CH}_2\text{O}] + \text{H}_2\text{O} + \text{O}_2$
 - ✓ General: $\text{CO}_2 + 2\text{H}_2\text{X} \rightarrow [\text{CH}_2\text{O}] + \text{H}_2\text{O} + \text{X}_2$
- ❖ Twenty years later, scientists confirmed van Niel's hypothesis by using ^{18}O , a heavy isotope, as a tracer.
- ❖ Hydrogen extracted from water is incorporated into sugar, and oxygen is released to the atmosphere.



Respiration and photosynthesis– a comparison

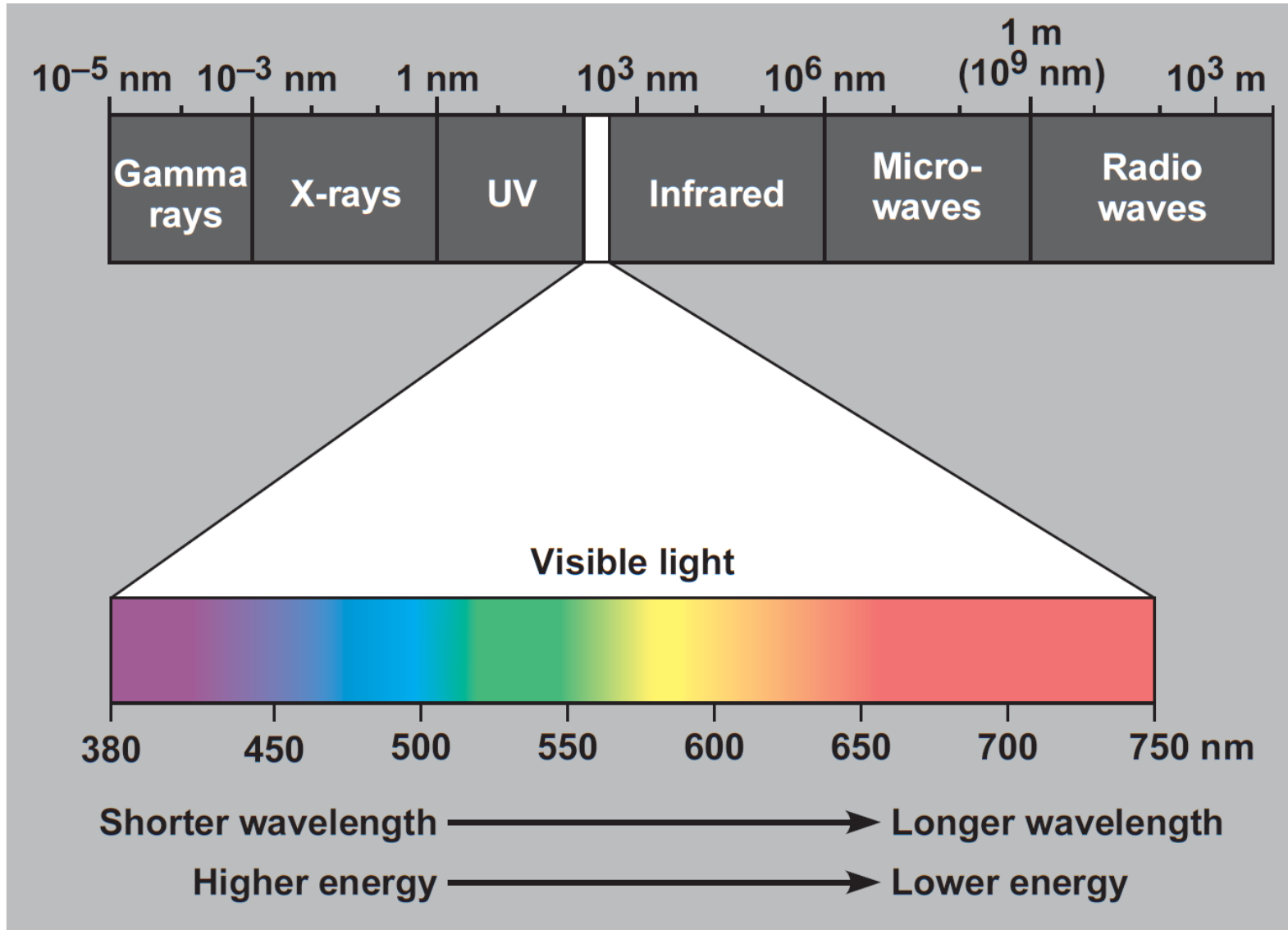
- ❖ **Both photosynthesis and aerobic respiration involve redox reactions.**
- ✓ *During **cellular respiration**, energy is released from sugar when **electrons associated with hydrogen** are transported by carriers to oxygen, forming water as a by-product.*
- ✓ *The electrons lose potential energy as they “fall” down the electron transport chain toward electronegative oxygen, and the mitochondrion harnesses that energy to synthesize ATP.*
- ❖ **Photosynthesis reverses the direction of electron flow.**
- ✓ Water is split and electrons are transferred with H^+ from water to CO_2 , reducing it to sugar.
- ✓ Because the electrons increase in potential energy as they move from water to sugar, the process requires energy. **The energy boost is provided by light.**

Photosynthesis- Two stages



- ❖ Photosynthesis is two processes, each with multiple steps: **light reactions** and the **Calvin cycle (dark reactions)**.
 - ✓ The **light reactions** (*photo*) convert solar energy to chemical energy.
 - ✓ The **Calvin cycle** (*synthesis*) uses energy from the light reactions to incorporate CO_2 from the atmosphere into sugar.
- ❖ **Light reactions occur at the thylakoids, the Calvin cycle occurs in the stroma.**
 - ✓ In the thylakoids, molecules of $NADP^+$ and ADP pick up electrons and phosphate, respectively, and $NADPH$ and ATP are then released to the stroma, where they play crucial roles in the Calvin cycle.

Electromagnetic spectrum



$$E = Nh\nu = \frac{Nhc}{\lambda}$$

N = Avagardo's number = 6.023×10^{23} photons/einstein

h = Planck's constant = 6.627×10^{-27} erg sec
= 1.58×10^{-34} cal sec

ν = frequency of the light 1/sec.

The energy in one Einstein (one mole of photons) is

$$E = Nh\nu$$
$$= \frac{2.855}{\lambda} \text{ cal/einstein}$$

Visible light drives photosynthesis

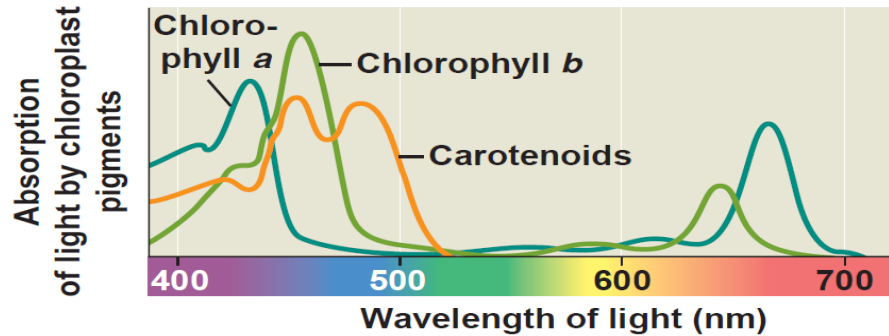


- ❖ Light is a form of electromagnetic energy or radiation and travels in rhythmic waves.
- ❖ The distance between crests of electromagnetic waves is called the **wavelength**.
 - ✓ Wavelengths of electromagnetic radiation range from shorter than a nanometer (gamma rays) to longer than a kilometer (radio waves).
- ❖ The entire range of electromagnetic radiation is the **electromagnetic spectrum**.
- ❖ **The most important segment of the electromagnetic spectrum for life is a narrow band between 380 and 750 nm, the band of visible light detected as colors by the human eye.**
- ❖ Although light travels as a wave, many of its properties are those of a discrete particle, a **photon**.
- ❖ Visible light is the radiation that drives photosynthesis.

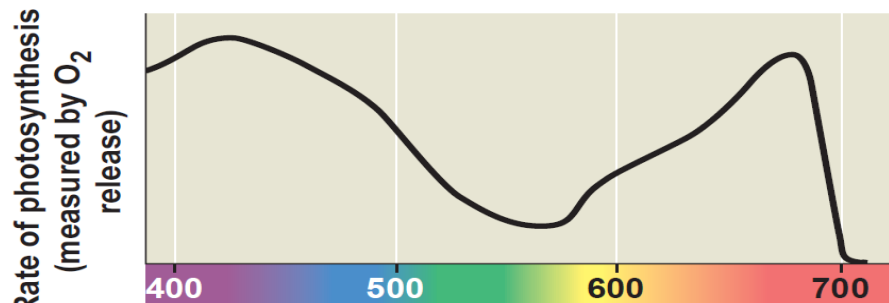
Photosynthetic pigments are light receptors



- ❖ When light meets matter, it may be reflected, transmitted, or absorbed. Substances that absorb visible light are known as *pigments*.
- ❖ There are three important pigments in plant: (a) Chlorophyll a (b) Chlorophyll b (c) Carotenoids, the accessory pigments.



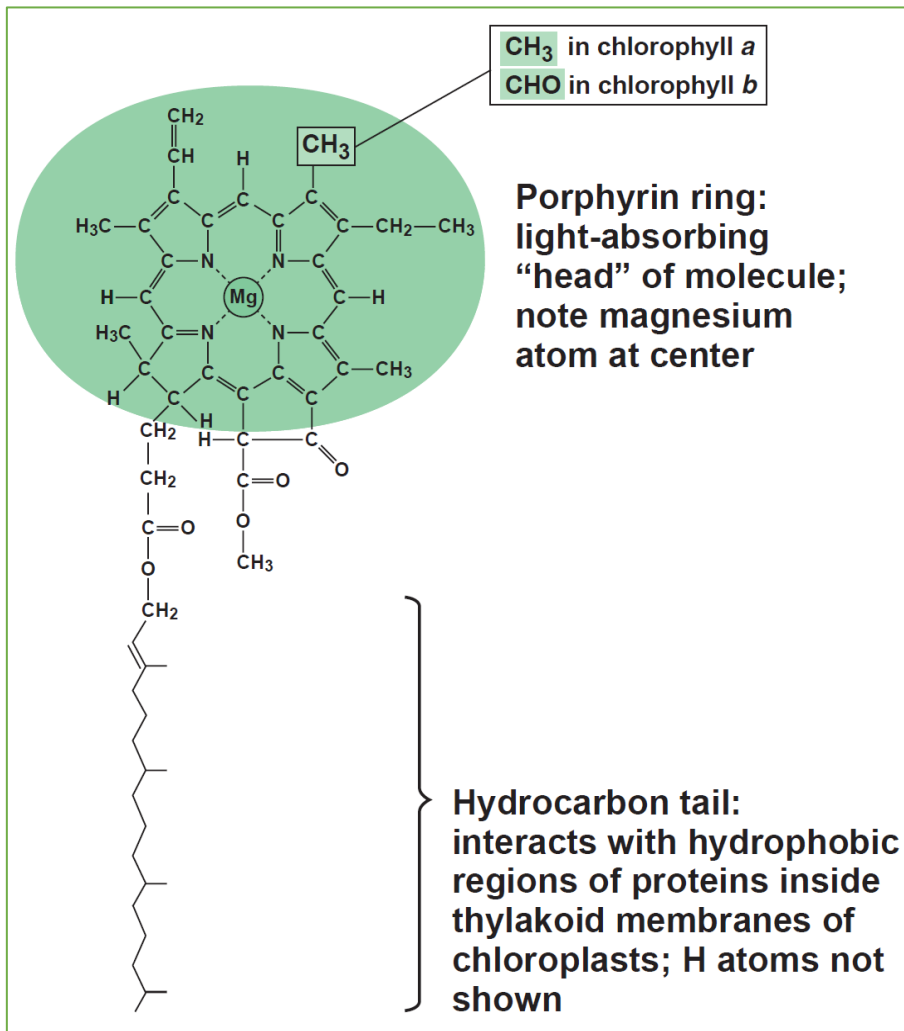
(a) Absorption spectra



(b) Action spectrum

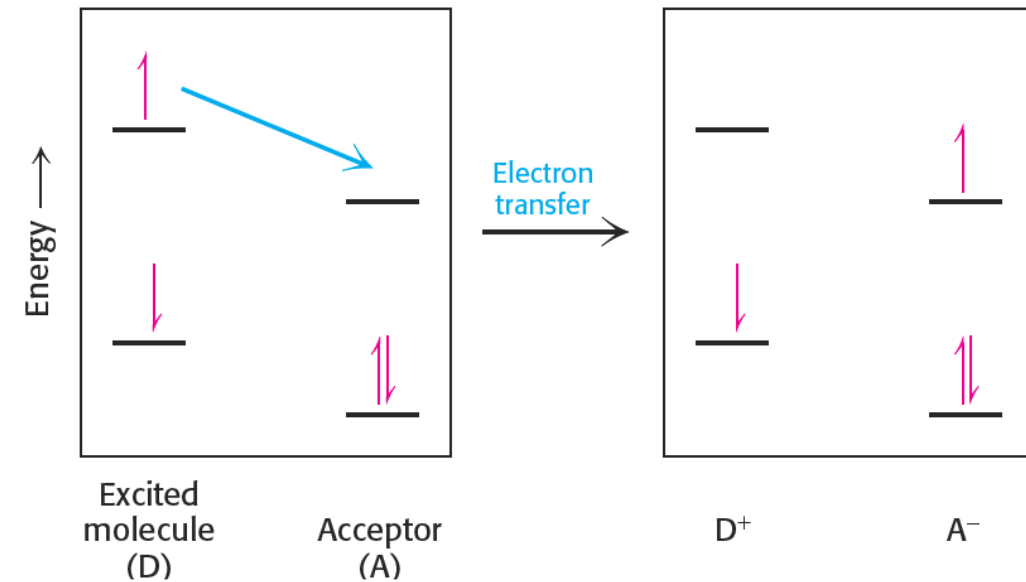
- ❖ The spectrum of chlorophyll *a* suggests that violet-blue and red light work best for photosynthesis, since they are absorbed, while green is the least effective color.
- ❖ This is confirmed by an **action spectrum** for photosynthesis.
- ❖ An action spectrum measures changes in some measure of photosynthetic activity (for example, O₂ release) as the wavelength is varied.

Chlorophyll a, b, and caratenoids

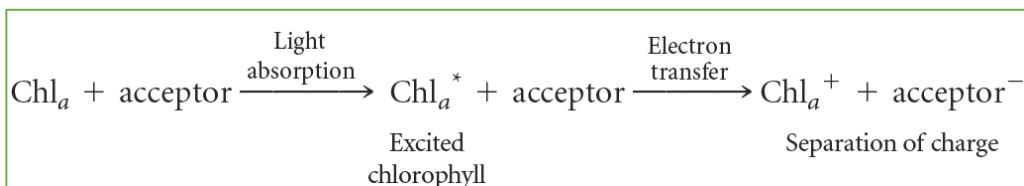


- ❖ Only chlorophyll *a* participates directly in the light reactions, but accessory photosynthetic pigments absorb light and transfer energy to chlorophyll *a*.
 - ✓ **Chlorophyll *b***, with a slightly different structure than chlorophyll *a*, has a slightly different absorption spectrum and *funnels the energy from these wavelengths* to chlorophyll *a*.
 - ✓ **Carotenoids** can funnel the energy from other wavelengths to chlorophyll *a* and also participate in photoprotection against excessive light.
 - ✓ These compounds absorb and dissipate excessive light energy that would otherwise damage chlorophyll or interact with oxygen to form reactive oxidative molecules that could damage the cell.

Photon, electron, and excitation

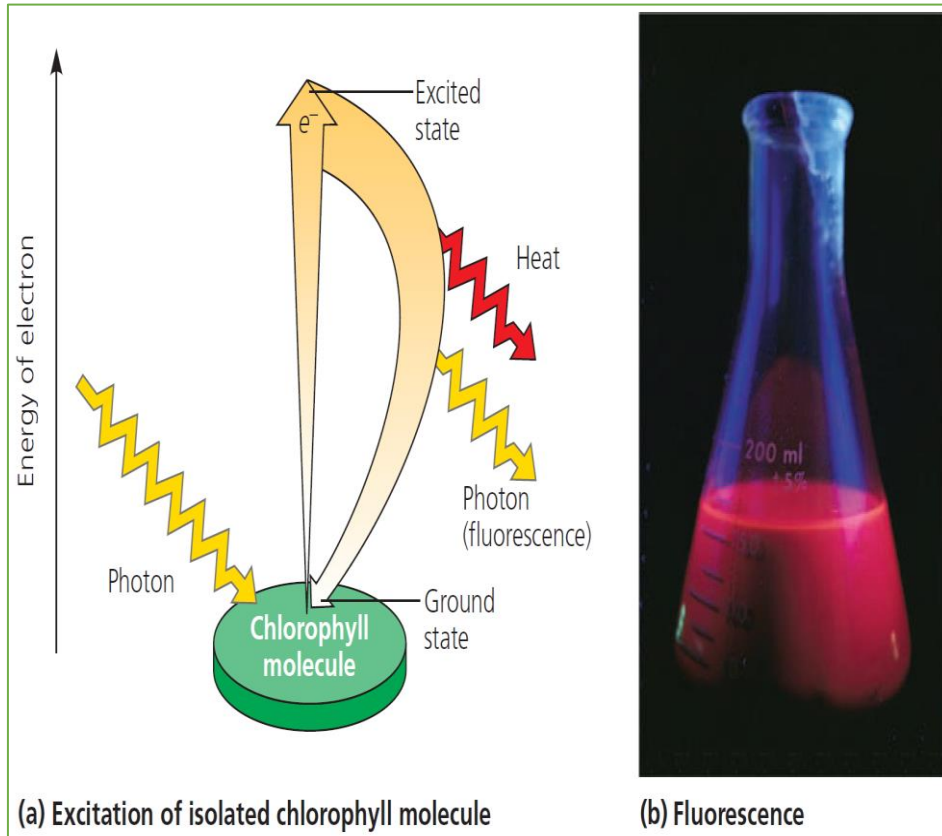


If a suitable electron acceptor is nearby, an electron that has been moved to a high energy level by light absorption can move from the excited molecule to the acceptor.



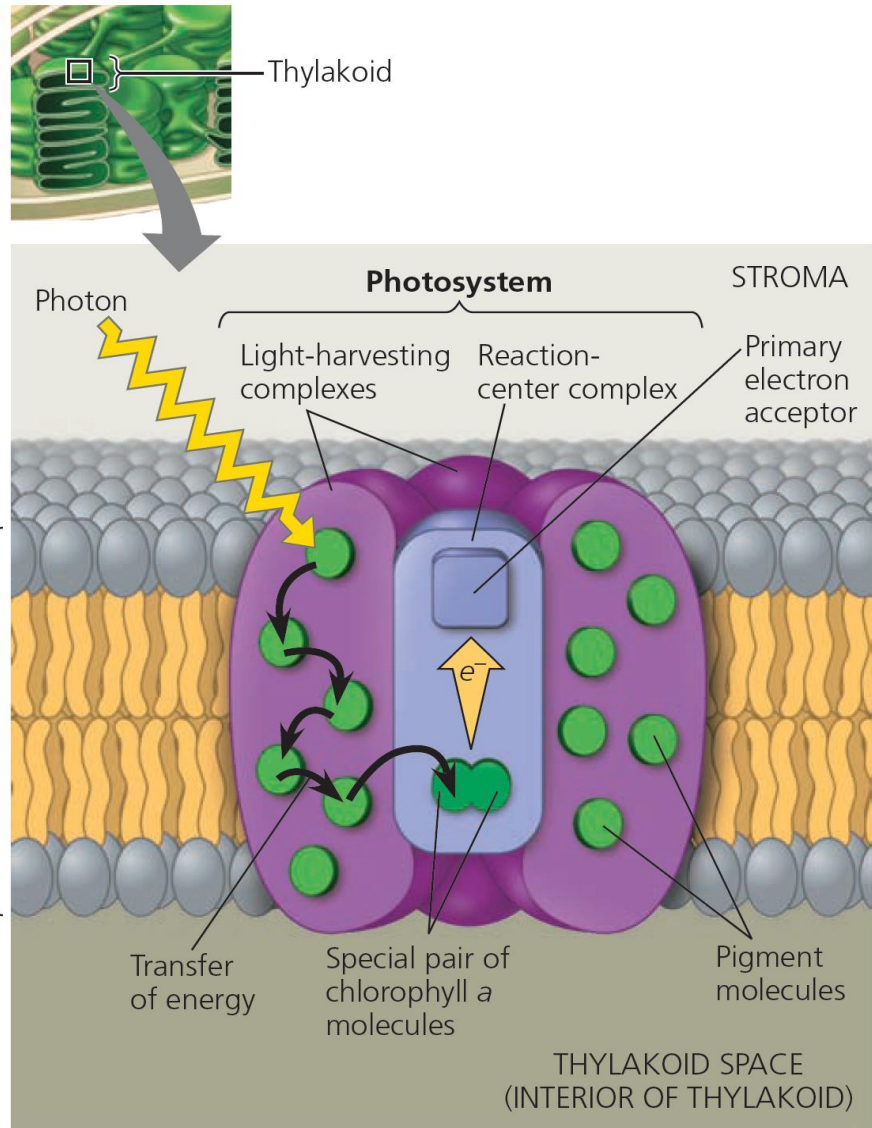
- ❖ When a molecule absorbs a photon, one of the molecule's electrons is elevated to an orbital with more potential energy.
 - ❖ The electron moves from its ground state to an excited state.
 - ❖ The only photons that a molecule can absorb are those whose energy matches exactly the energy difference between the ground state and the excited state of this electron.
 - ❖ Because this energy difference varies among atoms and molecules, a particular compound absorbs only photons corresponding to specific wavelengths.
- ✓ *This is the reason each pigment has a unique absorption spectrum.*

Photons, electrons, and fluorescence



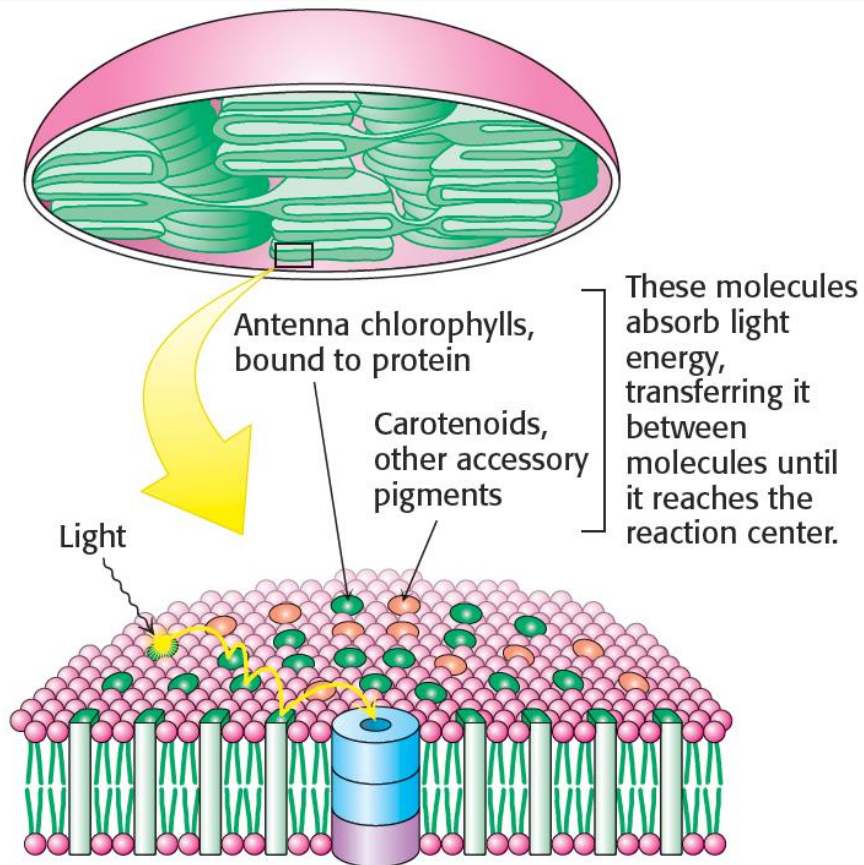
- ❖ Excited electrons are unstable. Generally, they drop to their ground state in a billionth of a second, releasing heat energy.
- ❖ In isolation, some pigments emit light after absorbing photons, in a process called **fluorescence**.
 - ✓ *If a solution of chlorophyll isolated from chloroplasts is illuminated, it fluoresces in the red-orange part of the spectrum and gives off heat.*
- ❖ Chlorophyll excited by absorption of **light energy produces very different results in an intact chloroplast than it does in isolation**.
- ❖ In the **thylakoid membrane**, chlorophyll is organized along with proteins and smaller organic molecules into photosystems.

Reaction-center and light-harvesting complexes

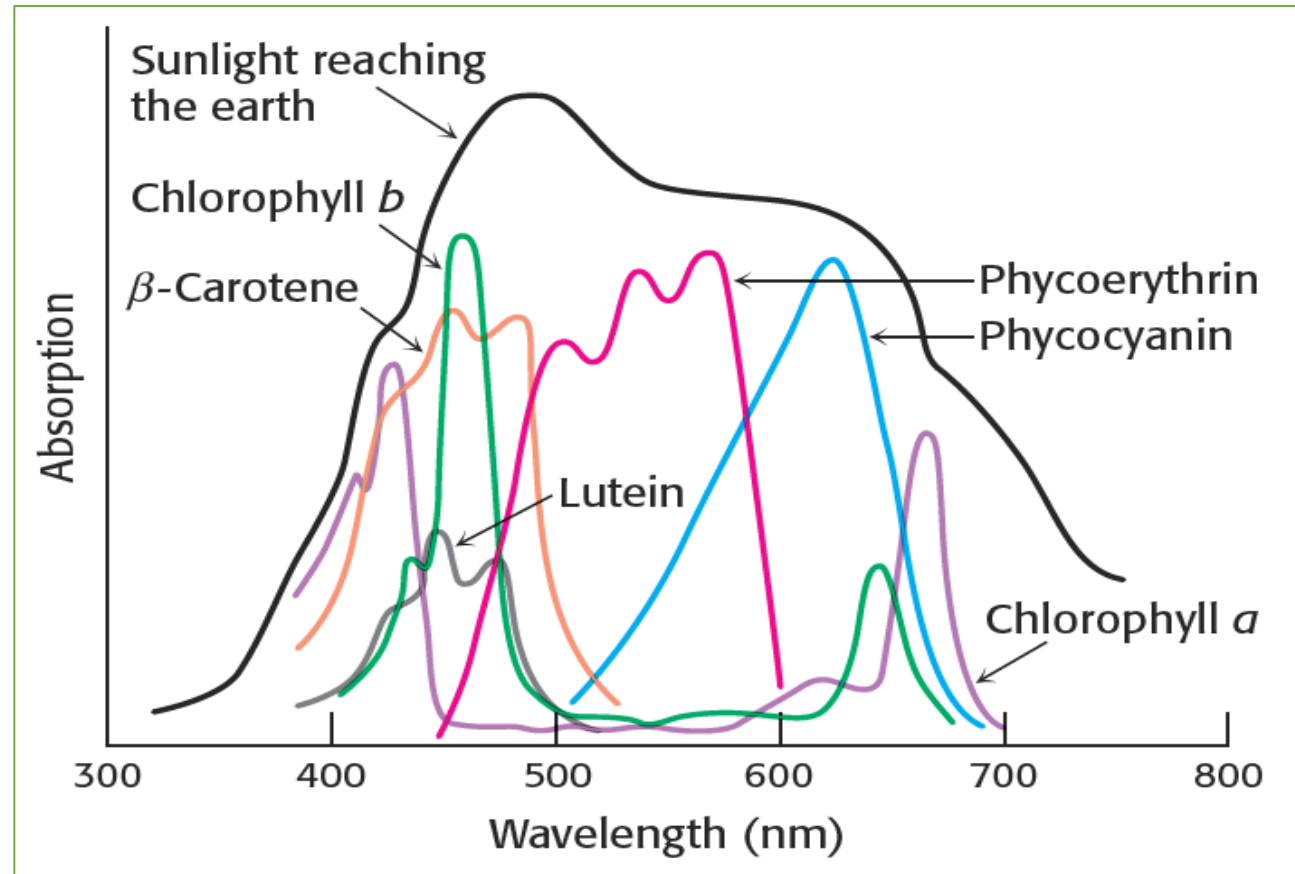


- ❖ In the thylakoid membrane, chlorophyll molecules are organized into clusters called photosystems.
- ❖ A **photosystem** contains two kinds of complexes:
 - ❖ A **reaction-center** complex surrounded by a number of light-harvesting complexes.
 - ❖ A **light-harvesting complex** consists of various pigment molecules bound to proteins.
- ❖ When a pigment molecule absorbs a photon, the energy is transferred from molecule to molecule, somewhat like a human “wave” at a sporting event, until it is passed into the reaction-center complex.
- ❖ The reaction-center complex contains a pair of special chlorophyll *a* molecules and a molecule called the primary electron acceptor, which, as its name indicates, is capable of accepting electrons and becoming reduced.

Continued..



Photochemical reaction here converts the energy of a photon into a separation of charge, initiating electron flow.



(A) Light energy absorbed by accessory chlorophyll molecules or other pigments can be transferred to reaction centers by resonance energy transfer (yellow arrows), where it drives photoinduced charge separation. (B) Accessory pigments increase the range of light absorption well beyond what is possible with chlorophyll *a* and *b* alone. [After D. L. Nelson and M. M. Cox, *Lehninger Principles of Biochemistry*, 5th ed. [W. H. Freeman and Company, 2008], (A) Fig. 19.52, (B) Fig. 19.48.]

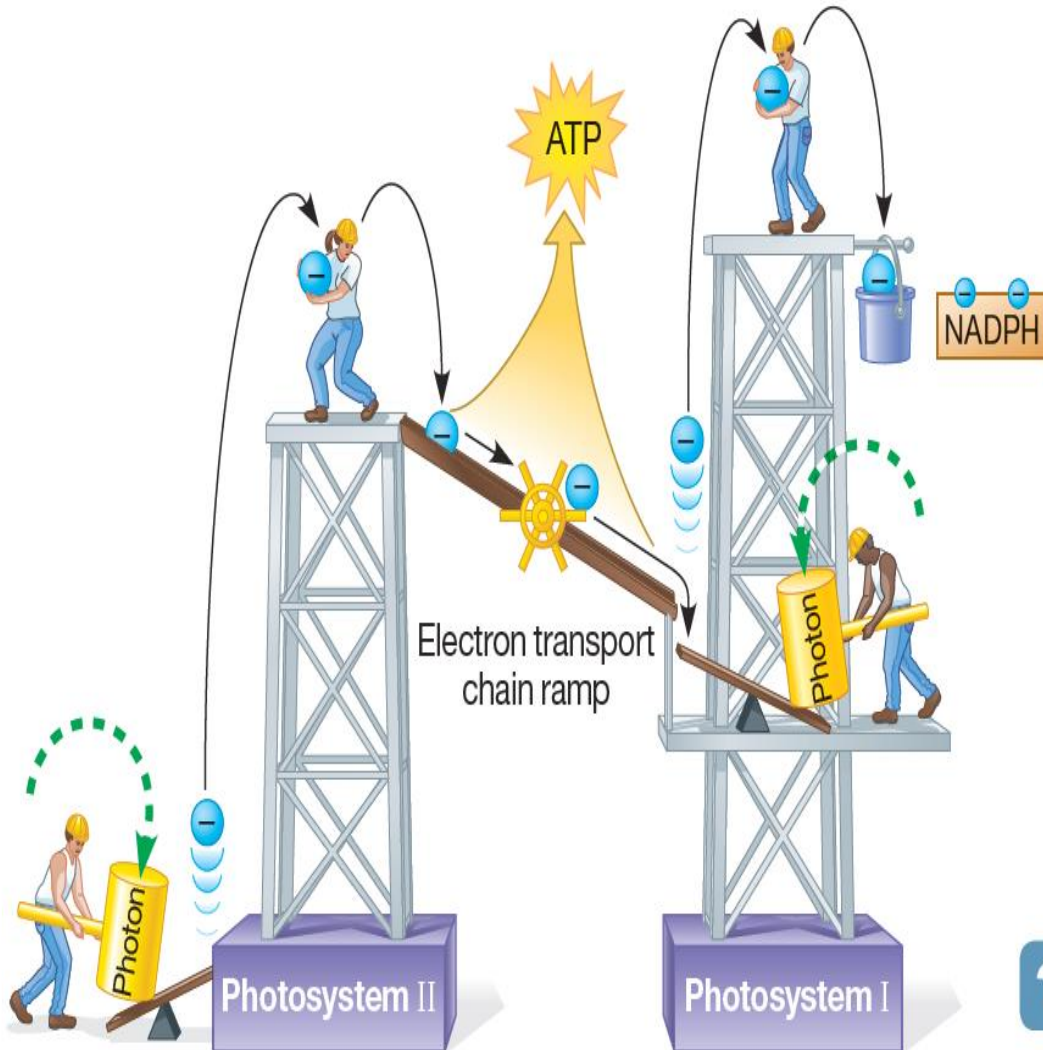
Photosystems-I and II



- ❖ **Photosystem II (PS II)** and **photosystem I (PS I)** each have a characteristic reaction-center complex—a particular kind of primary electron acceptor next to a pair of special chlorophyll *a* molecules associated with specific proteins.
 - ✓ The two photosystems were named in order of their discovery, but they function sequentially, with photosystem II functioning first.
- ❖ Photosystem II has a reaction-center **chlorophyll *a* known as P680**, with an absorption peak at 680 nm.
- ❖ Photosystem I has a reaction-center **chlorophyll *a* known as P700**, with an absorption peak at 700 nm.
 - ✓ These two pigments, P680 and P700, are nearly identical chlorophyll *a* molecules.
- ❖ Their **association with different proteins in the thylakoid membrane** affects the electron distribution in the chlorophyll molecules and accounts for the **slight differences in light-absorbing properties**.
- ❖ **These two photosystems work together in using light energy to generate ATP and NADPH.**

How Photosystems work together?

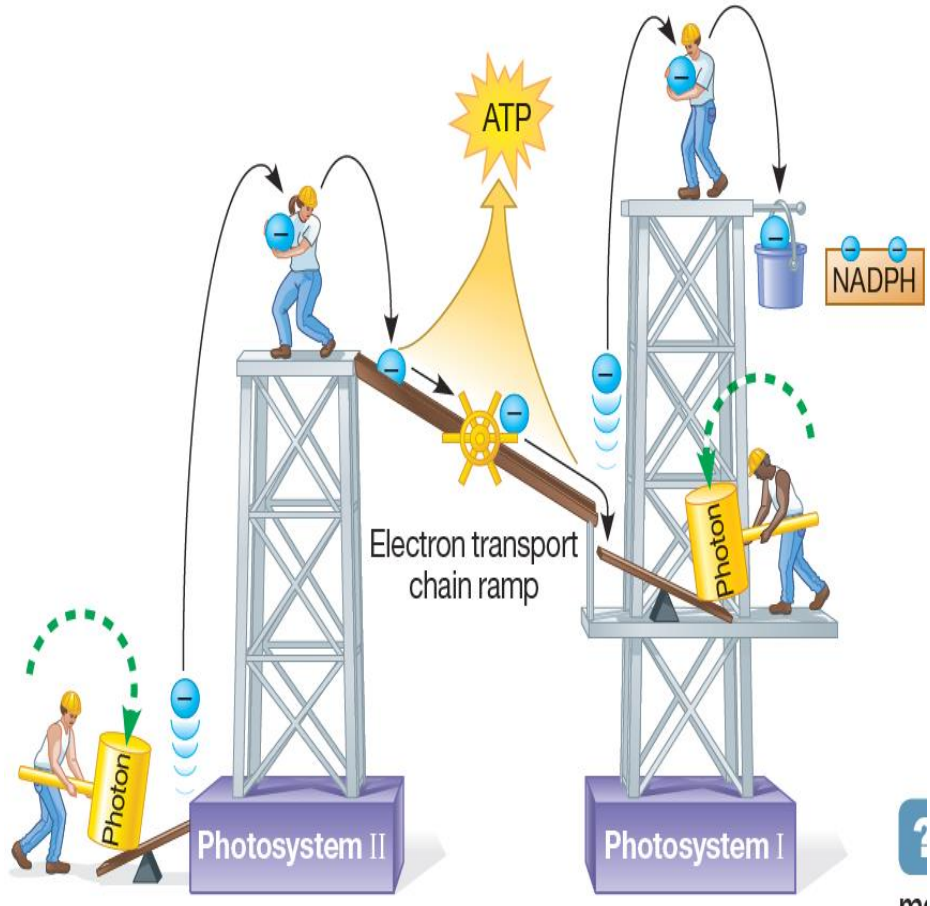
Mechanical analogy



- ❖ A large yellow photon mallet on the left provides the energy to kick electron from photosystem II to a higher energy level.
- ❖ This electron is caught by the primary electron acceptor standing on the platform.
- ❖ The electron is loaded onto an electron transport chain “ramp” leading to photosystem I. (Recall that photosystem II precedes photosystem I in the light reactions.)
- ❖ As electrons roll down the ramp, they release energy that is used for the production of ATP.

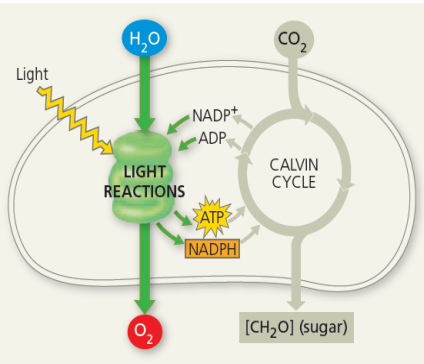
How Photosystems work together?

Mechanical analogy

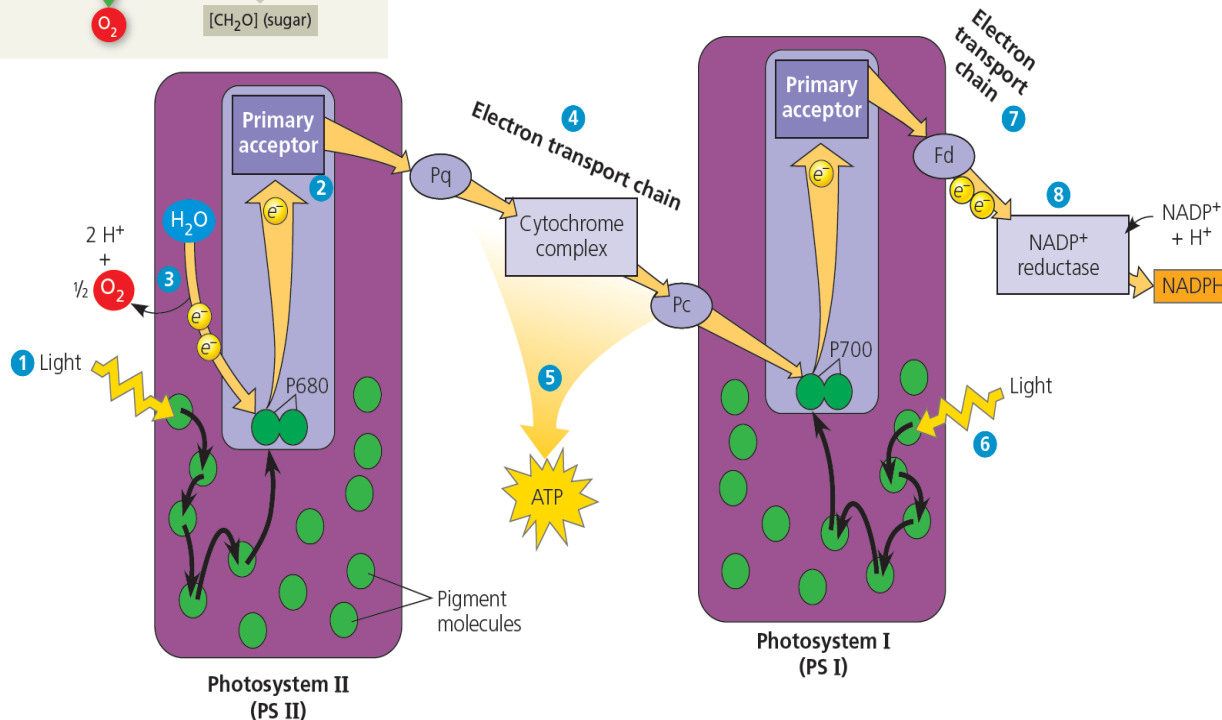


- ❖ When an electron reaches photosystem I, another photon mallet pumps it up to a higher energy level.
 - ❖ This electron is caught by a primary electron acceptor on the photosystem I platform.
 - ❖ From there, the photoexcited electrons are thrown into a bucket to produce NADPH.
 - ❖ This construction analogy shows how the coupling of two photosystems and an electron transport chain can transform the energy of light to the chemical energy of ATP and NADPH.
- *What is the source of the electrons that are moving through the photosystems to NADPH?*
 - *Don't the light reactions produce O_2 —where does that happen?*
 - *And how does the flow of electrons down that ramp produce ATP?*

ATP and NADPH production

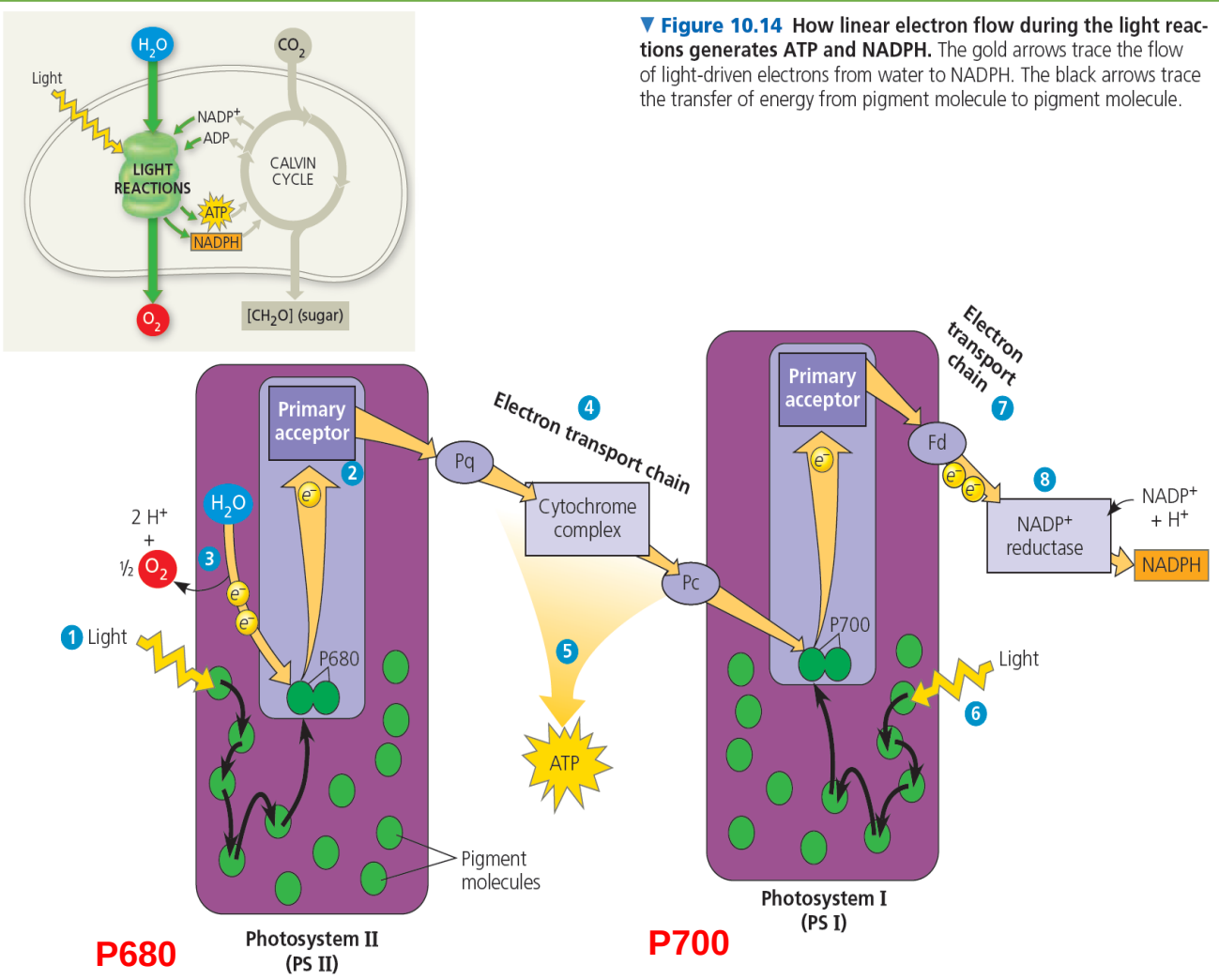


▼ **Figure 10.14** How linear electron flow during the light reactions generates ATP and NADPH. The gold arrows trace the flow of light-driven electrons from water to NADPH. The black arrows trace the transfer of energy from pigment molecule to pigment molecule.



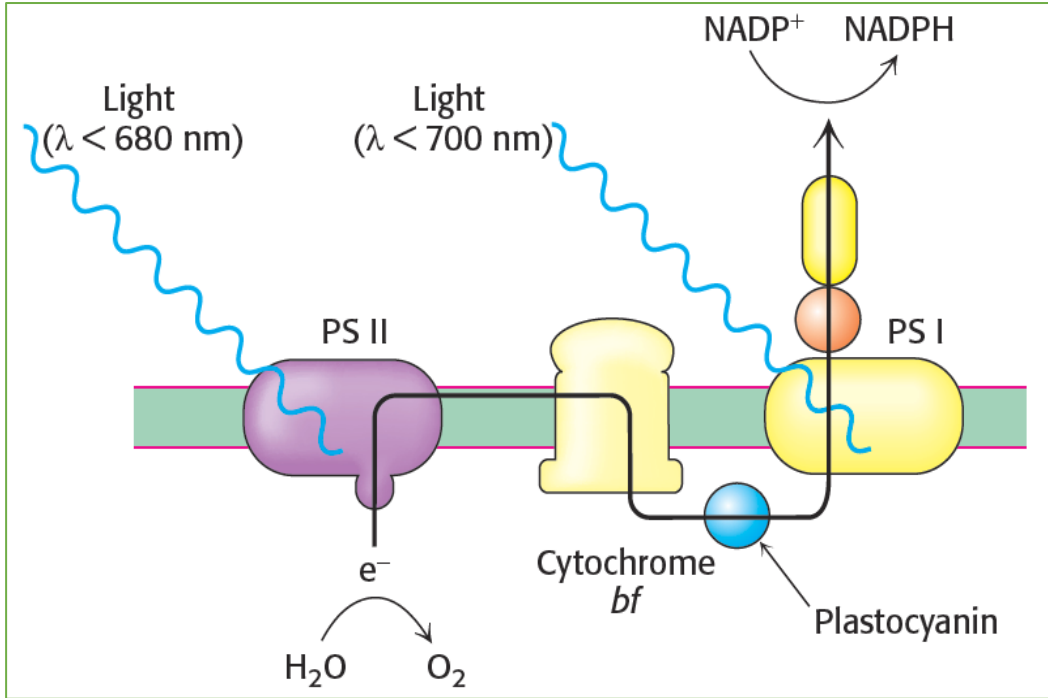
- ❖ The electrons that end up **reducing NADP⁺ to NADPH** originally come from water.
- ❖ An enzyme in the thylakoid space splits H₂O into 2 electrons, 2 hydrogen ions (H⁺), and 1 oxygen atom (1/2 O₂).
- ❖ The oxygen atom immediately joins with another oxygen to form O₂.
- ❖ Water is the source of the O₂ produced in photosynthesis, and these oxygen molecules diffuse out of the thylakoids, the chloroplast, and the plant cell, finally exiting the leaf through its stomata.

Summary of ATP and NADPH production



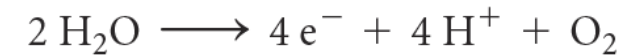
- ❖ The all-important electrons from water are passed, one by one, to the reaction center chlorophyll *a* molecules in photosystem II, replacing the photoexcited electron that was just captured by the primary electron acceptor.
- ❖ From photosystem II, the electrons pass through an electron transport chain to the reaction center chlorophyll *a* molecules in photosystem I, again replacing photoexcited electrons that had been captured by its primary electron acceptor.
- ❖ Although the illustration shows these electrons being dropped in a bucket, they actually are passed through a short electron transport chain to NADP+, reducing it to NADPH.

How photosystems work together? Summary



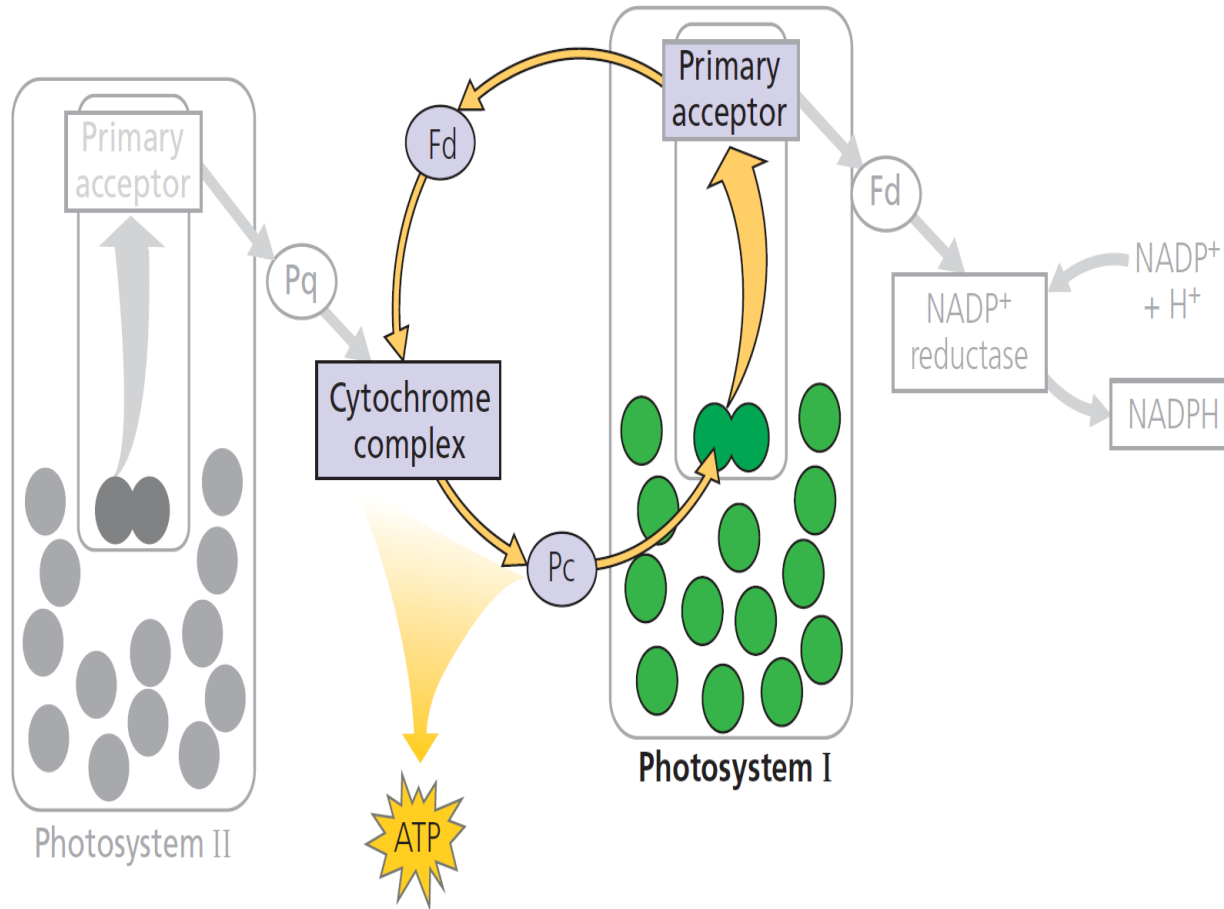
- ❖ Photosystem I responds to light with wavelengths shorter than 700 nm and is responsible for providing electrons to reduce NADP⁺ to NADPH, a versatile reagent for driving biosynthetic processes requiring reducing power.

- ❖ Photosystem II responds to wavelengths shorter than 680 nm, sending electrons through a membrane-bound proton pump called *cytochrome bf* and then on to photosystem I to replace the electrons donated by PS I to NADP⁺.
- ❖ The electrons in the reaction center of **photosystem II are replaced when two molecules of H₂O are oxidized to generate a molecule of O₂.**



- ❖ As we will soon see, *electrons flow from water through photosystem II, the cytochrome bf complex, and photosystem I and are finally accepted by NADP⁺.*
- ❖ In the course of this flow, a proton gradient is established across the thylakoid membrane. This proton gradient is the driving force for ATP production.

Two routes of electron flow–linear and cyclic

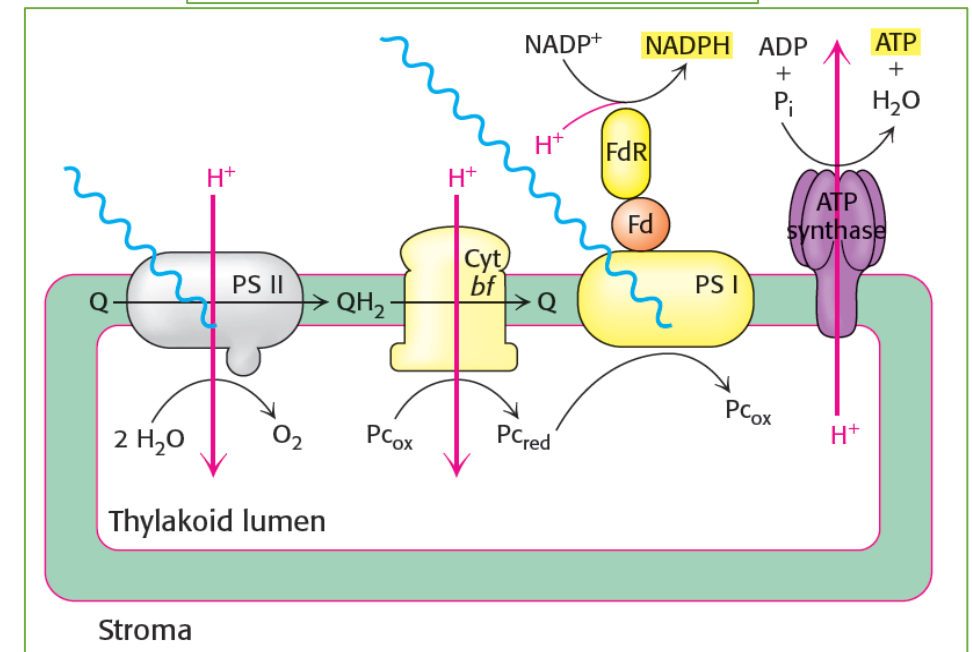
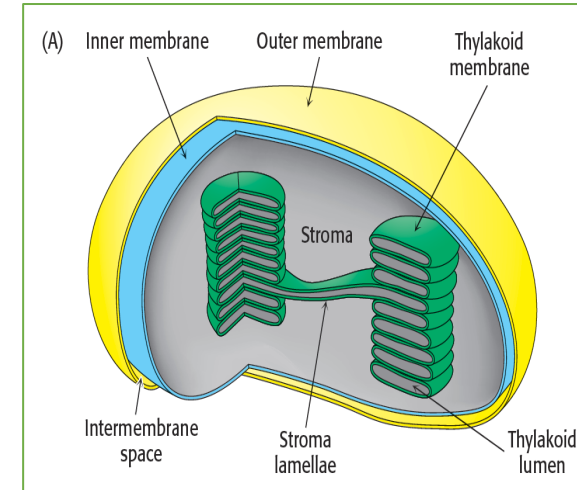


- ❖ Several living groups of photosynthetic bacteria have photosystem I but not photosystem II.
- ❖ In these species, which include the purple sulfur bacteria, **cyclic electron flow is the sole means of generating ATP in photosynthesis.**
 - ✓ Evolutionary biologists hypothesize that these bacterial groups are descendants of the bacteria in which photosynthesis first evolved, in a form similar to cyclic electron flow.

How proton gradient is generated?



- ❖ A crucial result of electron flow from water to NADP⁺ is the formation of a proton gradient. Such a gradient can be maintained because the thylakoid membrane is impermeable to protons.
- ❖ Energy inherent in the proton gradient, called the *proton-motive force* (Δp), is described as the sum of two components: a charge gradient and a chemical gradient. The dual nature of Δp is nicely illustrated in chloroplasts.
- ❖ In chloroplasts, nearly all of Δp arises from the proton gradient, whereas, in mitochondria, the contribution from the membrane potential is larger.
- ❖ The reason for this difference is that the thylakoid membrane, though impermeable to most ions, including H⁺, is quite permeable to Cl⁻ and Mg²⁺ acquired from soil.
- ❖ The light-induced transfer of H⁺ into the thylakoid lumen is accompanied by the transfer of either Cl⁻ in the same direction or Mg²⁺ (1 Mg²⁺ per 2 H⁺) in the opposite direction.
- ❖ Consequently, electrical neutrality is maintained and no membrane potential is generated.

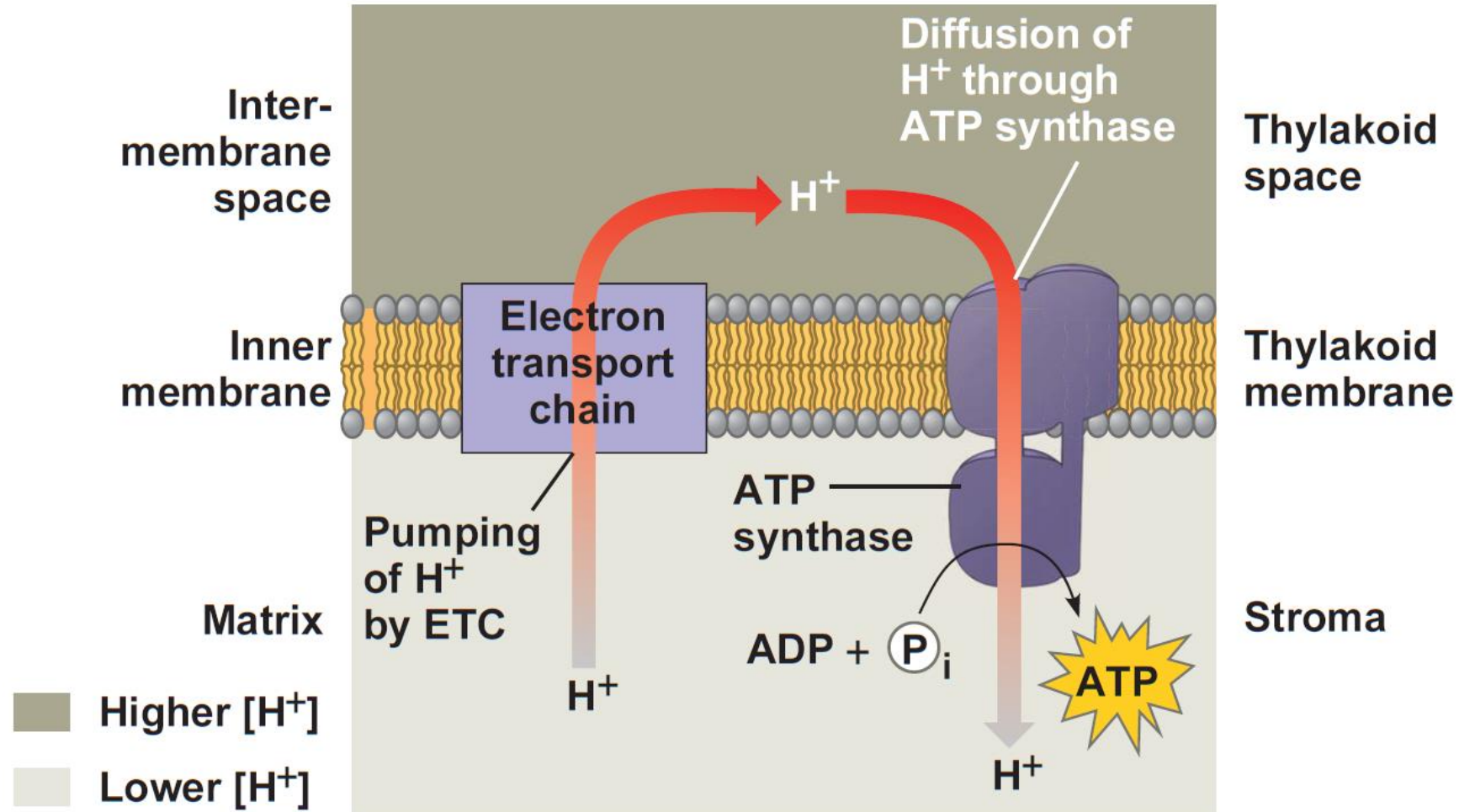


Comparison between Mito and chloroplast structure in ATP conversion

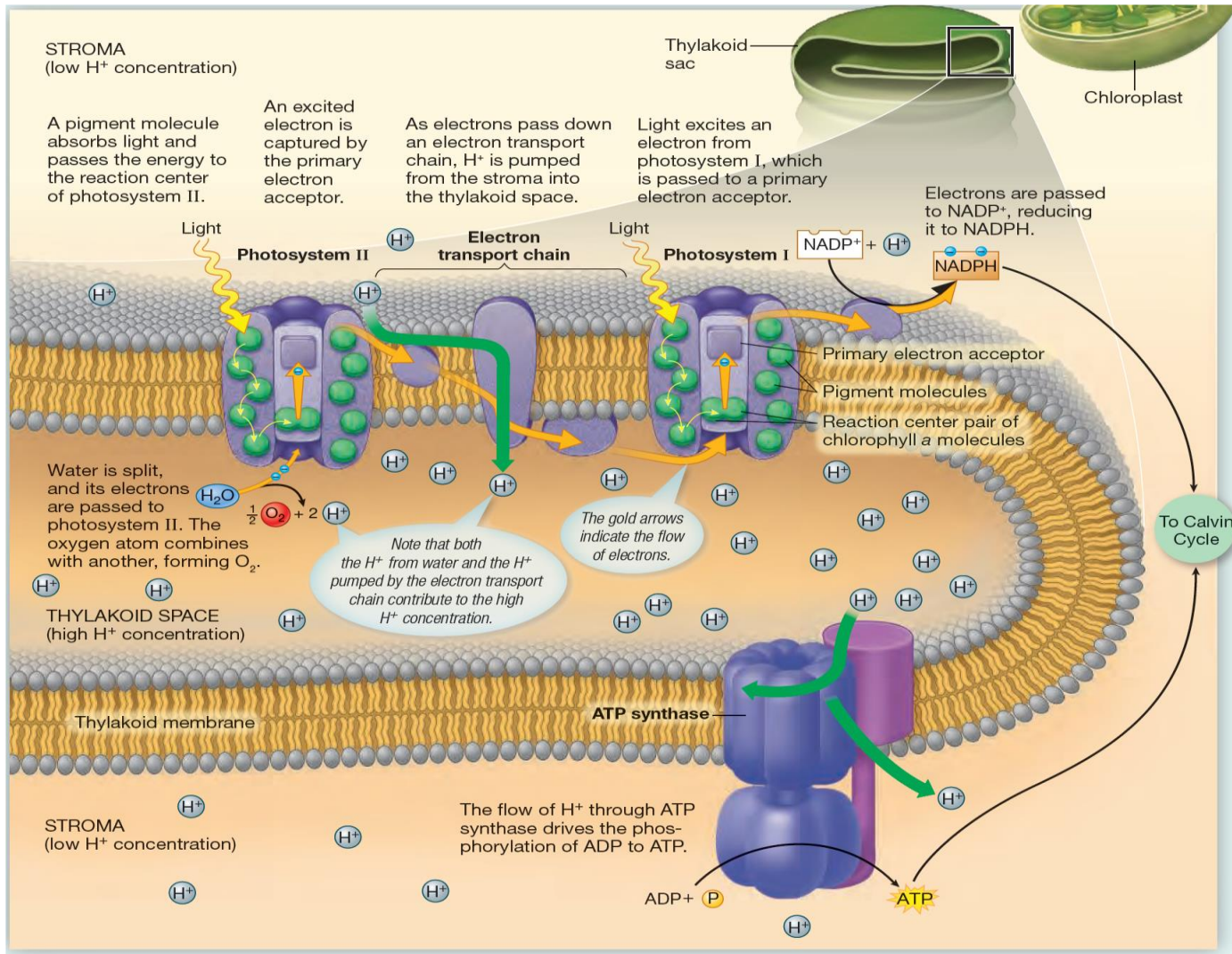


MITOCHONDRION STRUCTURE

CHLOROPLAST STRUCTURE



Summary of light reaction in photosynthesis



- ❖ Electron flow pushes electrons from water, where they have low potential energy, to NADPH, where they have high potential energy.
- ❖ The light-driven electron current also generates ATP.
- ❖ Thus, the equipment of the thylakoid membrane converts light energy to chemical energy stored in ATP and NADPH.
- ❖ This process also produces oxygen as a by-product.

Calvin Cycle



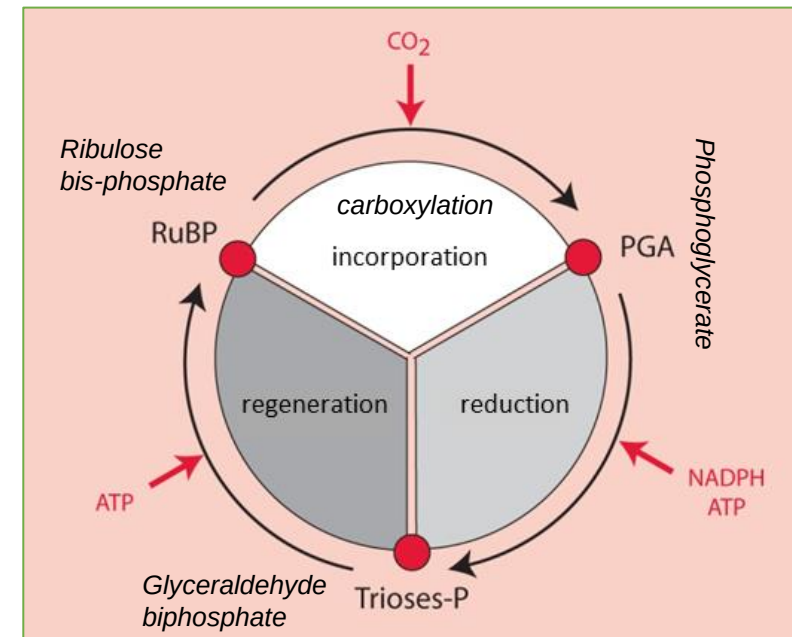
The Calvin cycle uses the chemical energy of ATP and NADPH to reduce CO₂ to sugar

- ❖ The Calvin cycle regenerates its starting material after molecules enter and leave the cycle.
- ❖ **The Calvin cycle is anabolic**, using energy to build sugar from smaller molecules.
- ❖ Carbon enters the cycle as CO₂ and leaves as sugar.
- ❖ The cycle spends the energy of ATP and the reducing power of electrons carried by NADPH to make sugar.
- ❖ **The actual sugar product of the Calvin cycle is not glucose but a three-carbon sugar, glyceraldehyde-3-phosphate (G3P).**

Calvin Cycle–Carbon fixation



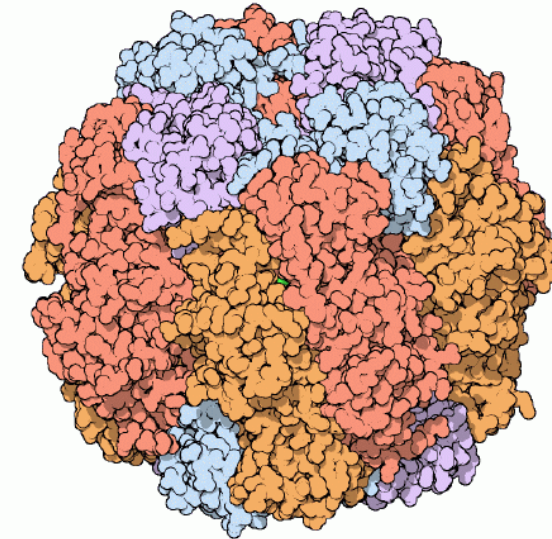
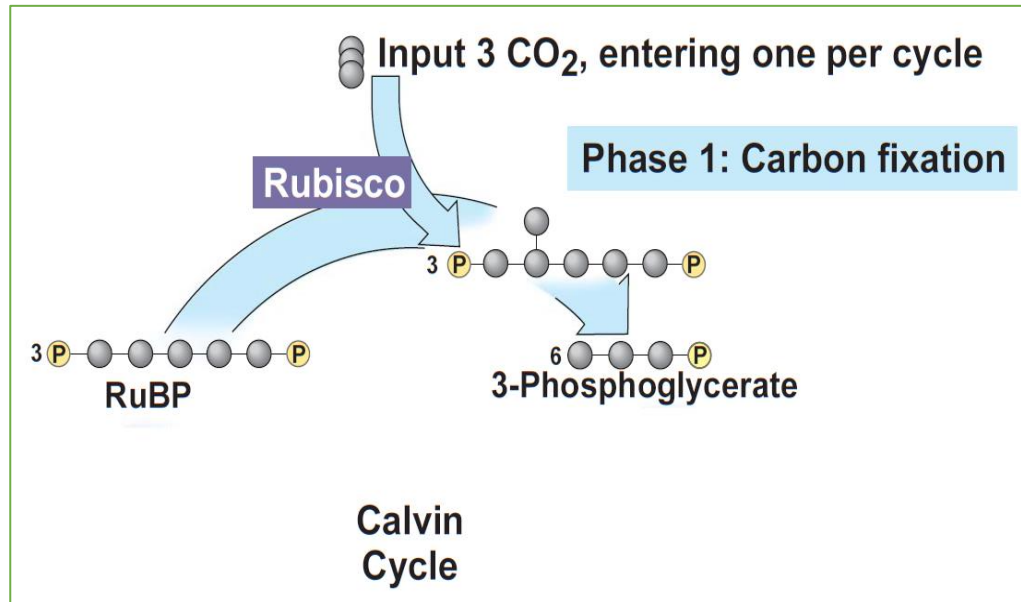
- ❖ Each turn of the Calvin cycle fixes one carbon.
- ❖ For the net synthesis of one G3P molecule, **the cycle must take place three times**, fixing three molecules of CO_2 .
- ❖ To make one glucose molecule requires six cycles and the fixation of six CO_2 molecules.
- ❖ The Calvin cycle has **three phases**:
 1. Carbon fixation,
 2. Reduction, and
 3. Regeneration of the CO_2 acceptor.



Calvin cycle-- Phase-I: Carbon fixation



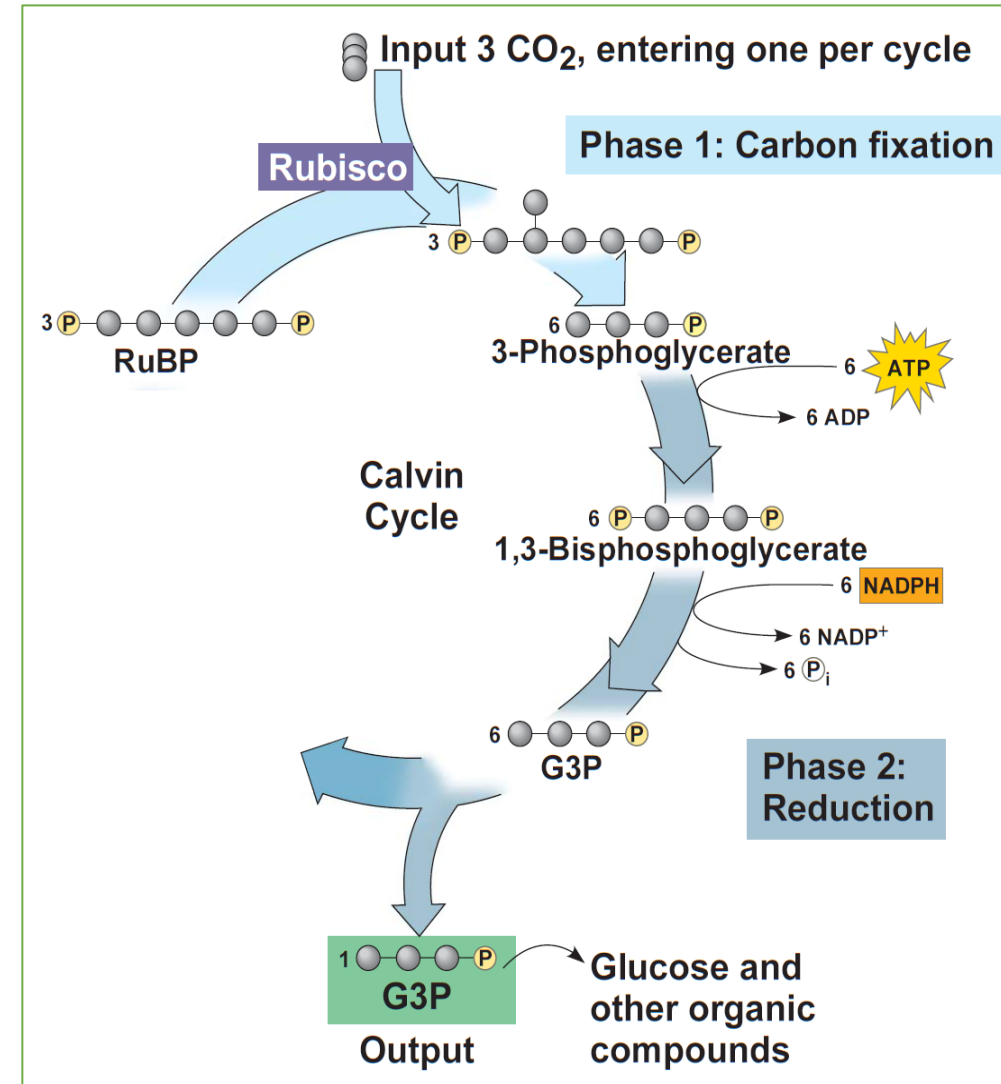
- ❖ In the **carbon fixation** phase, each CO_2 molecule is attached to a **five-carbon sugar, ribulose biphosphate (RuBP)**.
 - This reaction is catalyzed by **RuBP carboxylase-oxygenase, or rubisco**.
 - **Rubisco is the most abundant protein in chloroplasts and probably the most abundant protein on Earth.**
- ❖ The six-carbon intermediate is unstable and splits in half to form two molecules of 3-phosphoglycerate for each CO_2 fixed.



Calvin Cycle – Phase-II –Reduction



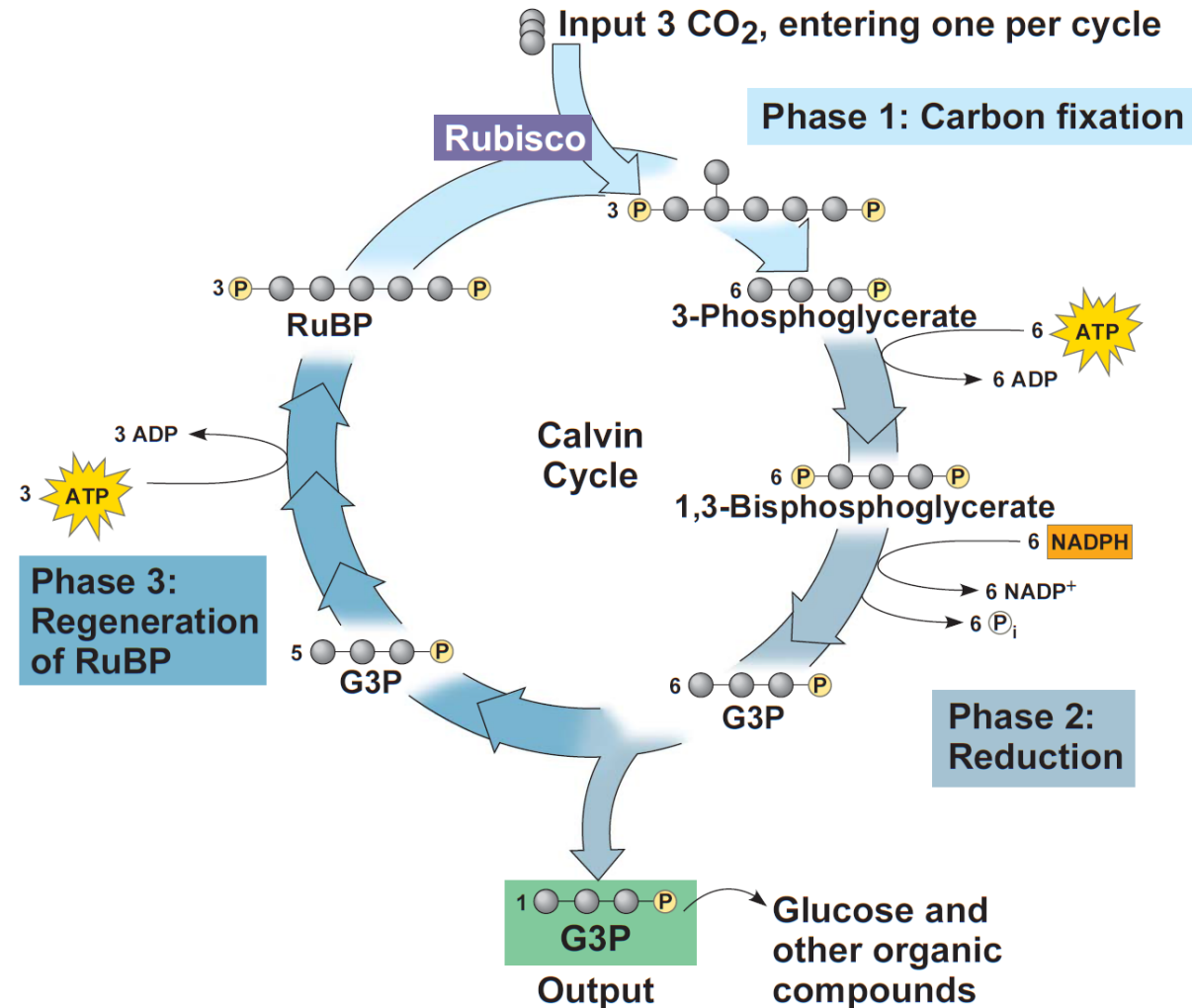
- ❖ During **reduction**, each 3-phosphoglycerate receives another phosphate group from ATP to form 1,3-bisphosphoglycerate.
- ❖ A pair of electrons from NADPH reduces each 1,3-bisphosphoglycerate to **G3P**.
- ❖ The electrons reduce a carboxyl group to the aldehyde group of G3P, which stores more potential energy.
- ❖ For every *three* molecules of CO_2 that enter the cycle, there are *six* molecules of G3P formed.
- ❖ One of these six G3P is a net gain of carbohydrate.
- ❖ This molecule (G3P) exits the cycle to be used by the plant cell, while the other five molecules are recycled to regenerate the three molecules of RuBP.



Calvin Cycle – Phase-III –Regeneration



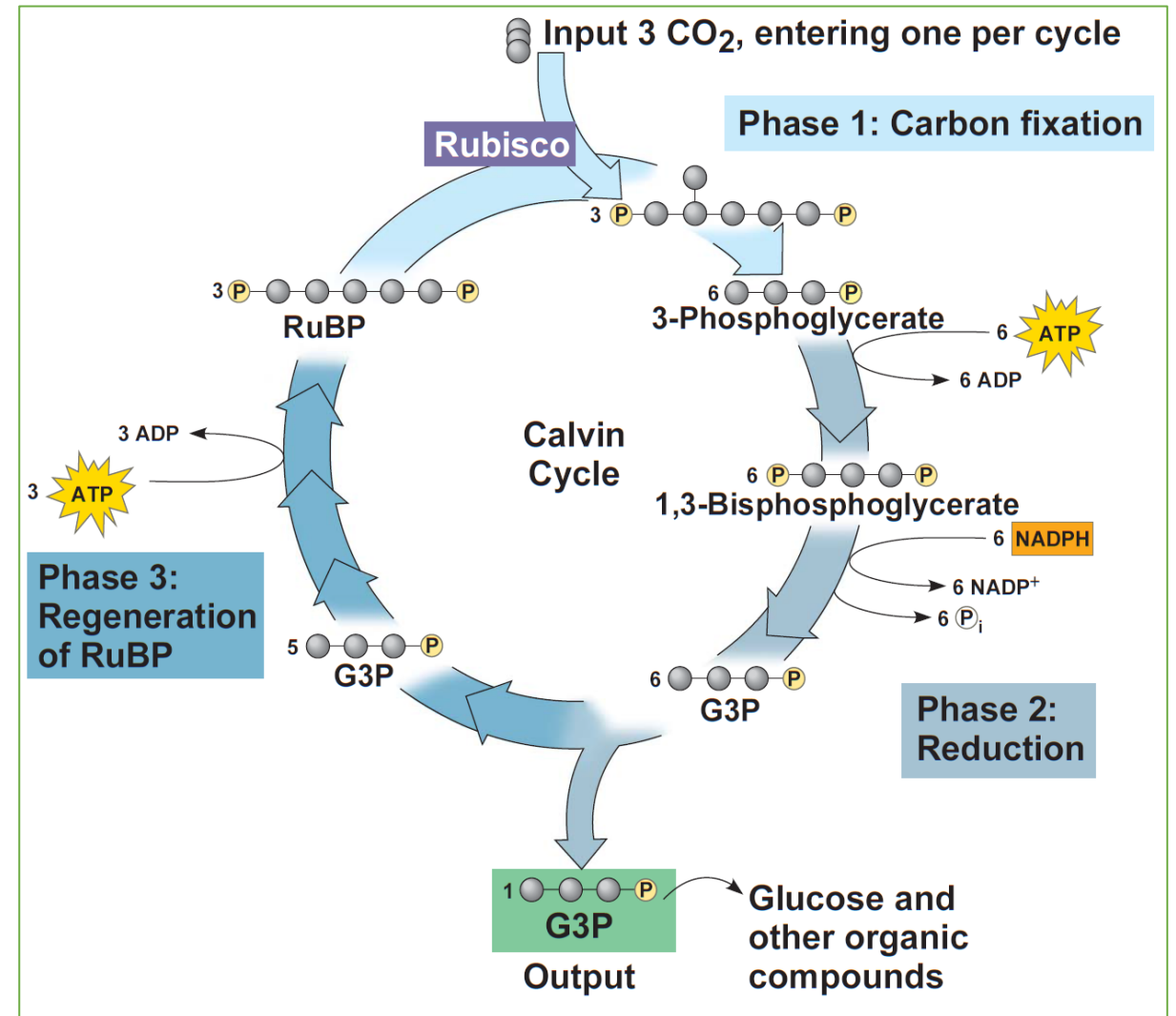
- ❖ In a complex series of reactions, the carbon skeletons of five molecules of G3P are rearranged by the last steps of the Calvin cycle to **regenerate three molecules of RuBP**.
- ❖ To accomplish this, the cycle spends three more molecules of ATP.
- ❖ The RuBP is now prepared to receive CO_2 again, and the cycle continues.



ATP and NADPH consumed



- ❖ For the net synthesis of one G3P molecule, the Calvin cycle consumes **nine ATP and six NADPH**.
- ❖ The light reactions regenerate ATP and NADPH.
- ❖ The G3P from the Calvin cycle is the starting material for metabolic pathways that synthesize other organic compounds, including glucose and other carbohydrates.
- ❖ Neither the light reactions nor the Calvin cycle alone can make sugar from CO₂. **Photosynthesis is an emergent property of the intact chloroplast that integrates the two stages of photosynthesis.**



Alternative mechanism– Photorespiration

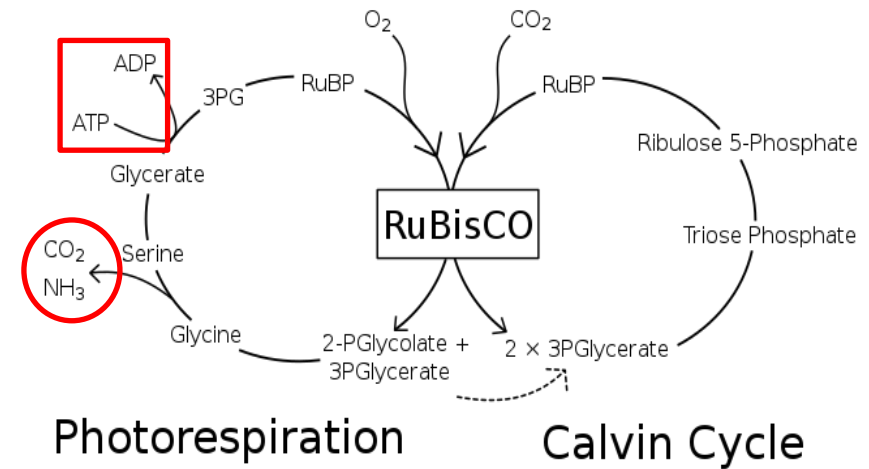
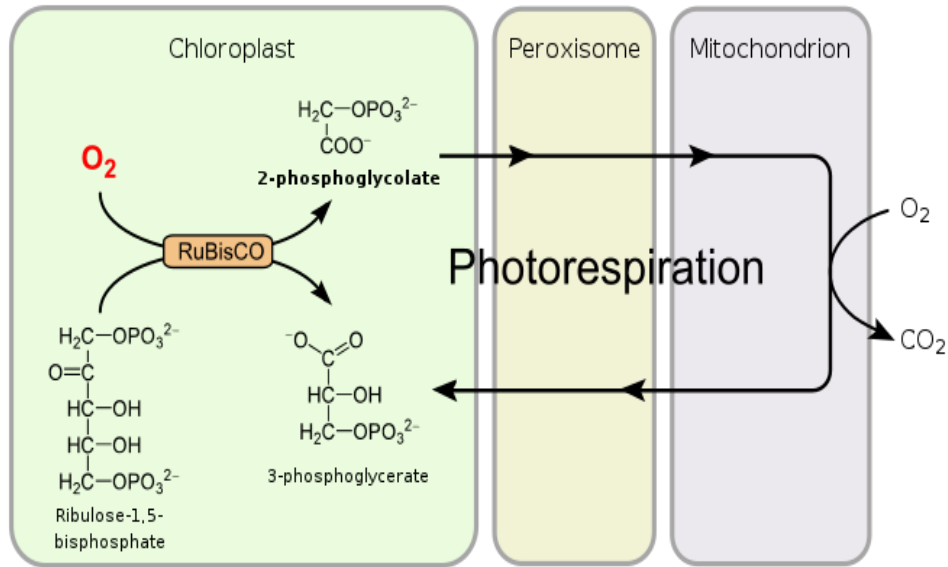


- ❖ One of the major problems facing terrestrial plants is **dehydration (loss of water)**.
- ❖ Metabolic **adaptations** to reduce dehydration often require trade-offs with other metabolic processes, especially photosynthesis.
- ❖ The **stomata are both the major route for gas exchange** (CO_2 in and O_2 out) and the main site of the evaporative loss of water.
 - ❖ ***On hot, dry days, plants close their stomata to conserve water.***
 - ❖ *With stomata closed, CO_2 concentrations in the air space within the leaf decrease and the concentration of O_2 released from the light reactions increases.*
 - ❖ *These conditions within the leaf favor an apparently wasteful process called **photorespiration**.*

Photorespiration–oxygen utilization by Rubisco

- ❖ In most plants (**C₃ plants**), initial fixation of CO₂ occurs via rubisco, forming a three-carbon compound, 3-phosphoglycerate.
 - ✓ **C₃ plants include rice, wheat, and soybeans.**
- ❖ When the stomata of C₃ plants partially close on a hot, dry day, CO₂ levels drop as CO₂ is consumed in the Calvin cycle.
- ❖ At the same time, O₂ levels rise as the light reactions convert light to chemical energy.
- ❖ **Although rubisco normally accepts CO₂, as CO₂ becomes scarce, rubisco can add O₂ to RuBP.**
- ❖ *When rubisco adds O₂ to RuBP, RuBP splits into a three-carbon compound and a two-carbon compound in a process called **photorespiration**.*
- ❖ The two-carbon compound is exported from the chloroplast and degraded to CO₂ by mitochondria and peroxisomes.
- ❖ Unlike normal respiration, this process produces no ATP. **In fact, photorespiration consumes ATP.**
- ❖ **Unlike photosynthesis, photorespiration does not produce sugar.** In fact, photorespiration decreases photosynthetic output by siphoning organic material from the Calvin cycle and releasing CO₂ that would otherwise be fixed.

Photorespiration– a toxic 2 phosphoglycolate is formed.



- ❖ The oxygenation reaction of RuBisCO is a wasteful process because [3-phosphoglycerate](#) is created at a lower rate and higher metabolic cost compared with [RuBP carboxylase activity](#).
- ❖ While photorespiratory carbon cycling results in the formation of [G3P](#) eventually, around 25% of carbon fixed by photorespiration is re-released as CO_2 [\[2\]](#) and nitrogen, as [ammonia](#).
- ❖ [Ammonia](#) must then be detoxified at a substantial cost to the cell.
- ❖ Photorespiration also incurs a direct cost of one [ATP](#) and one [NAD\(P\)H](#).

Photorespiration–Why is it important?

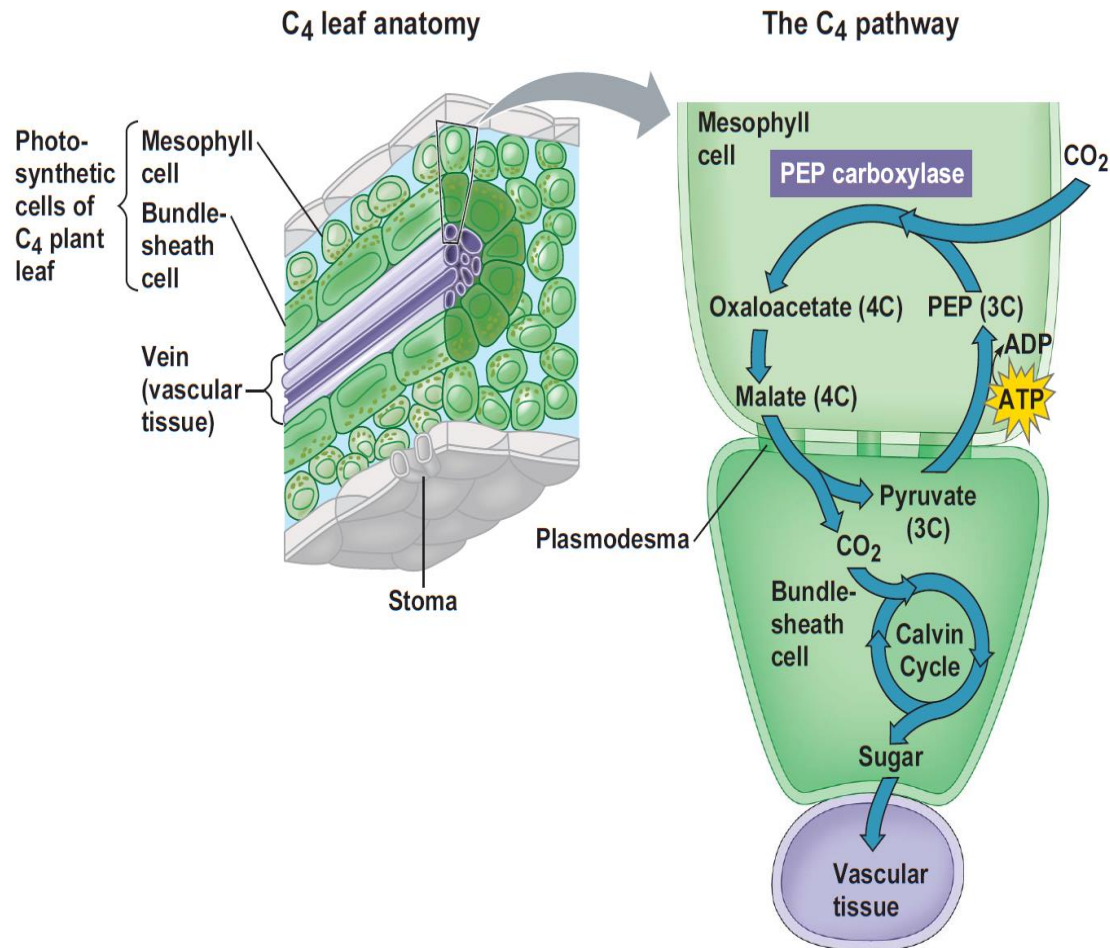


- ❖ At least in some cases, **photorespiration plays a protective role in plants.**
- ❖ Plants that are **genetically defective** in their ability to carry out photorespiration are **more susceptible to damage induced by excess light.**
- ❖ This is clear evidence that **photorespiration acts to neutralize otherwise damaging products of the light reactions**, which build up when a low CO₂ concentration limits the progress of the Calvin cycle.
- ❖ Whether there are other benefits of photorespiration is still unknown.
- ❖ **As heterotrophs dependent on carbon fixation in chloroplasts for food, humans naturally view photorespiration as wasteful.**
- ❖ **If photorespiration could be reduced in certain plant species without otherwise affecting photosynthetic productivity, crop yields and food supplies might increase.**

Optimization of Calvin cycle



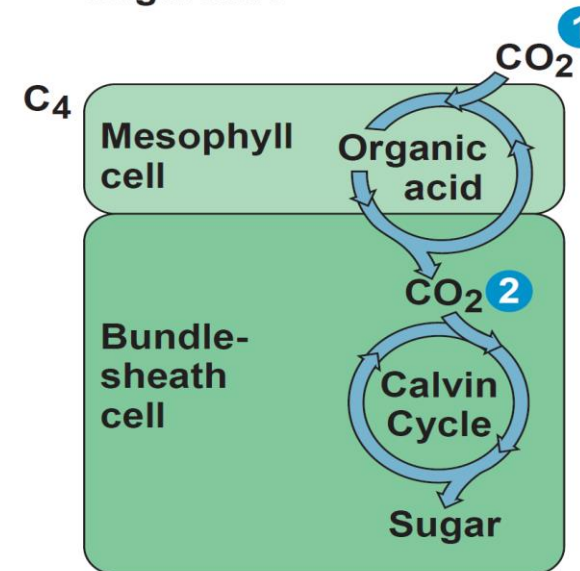
In some plant species, alternate modes of carbon fixation have evolved that minimize photorespiration and optimize the Calvin cycle—even in hot, arid climates.



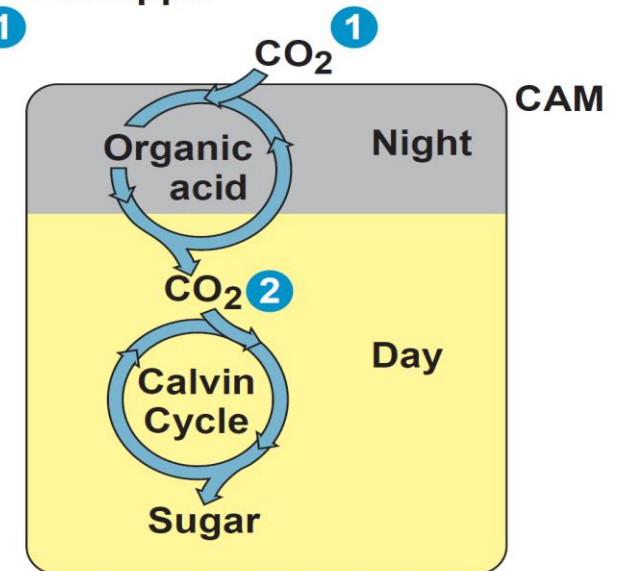
Sugarcane



Pineapple



(a) Spatial separation of steps



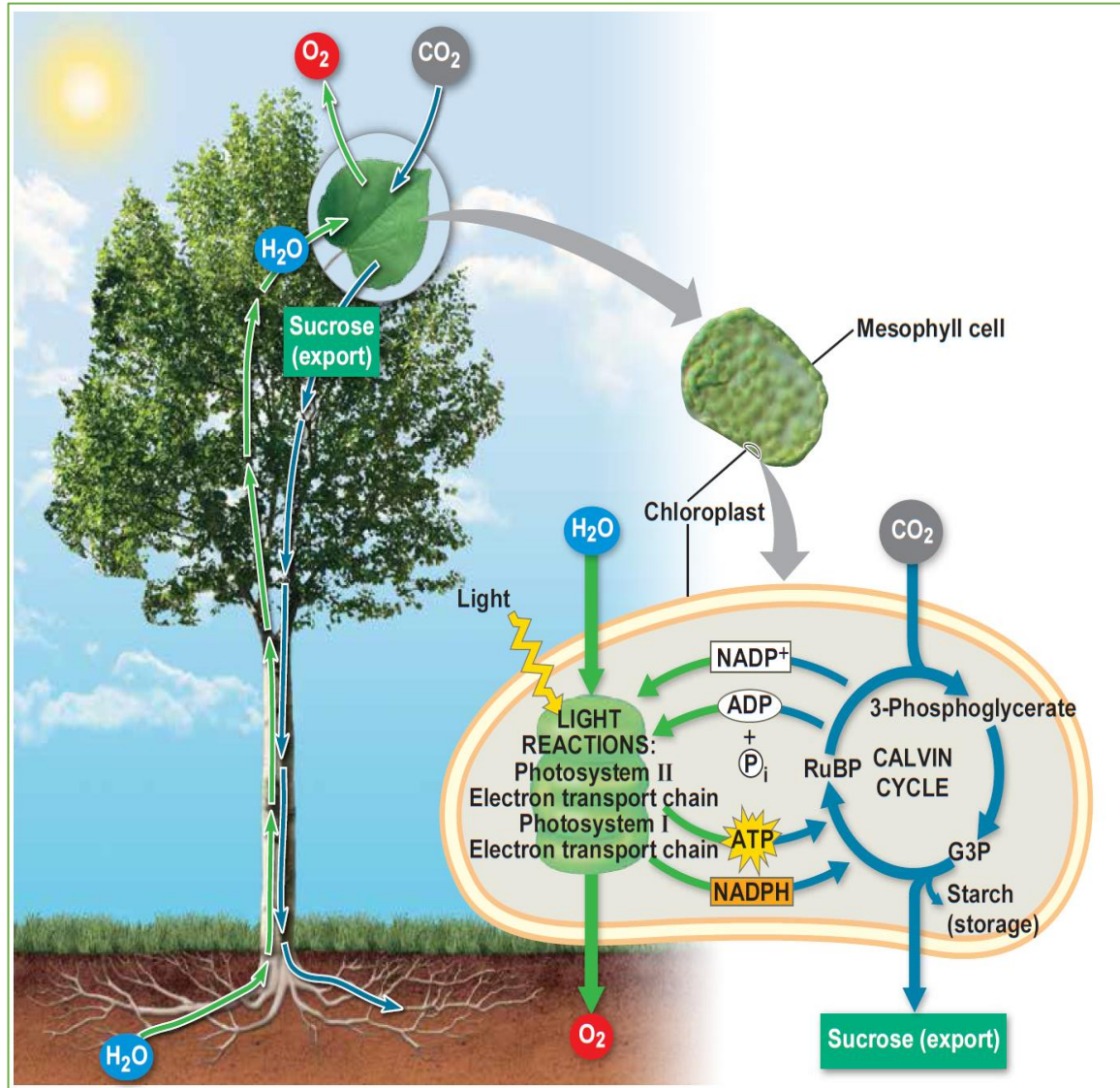
(b) Temporal separation of steps

Carbon fixation in C₄ plants



- ❖ C₄ pathway of carbon fixation is adapted by plants found in a dry tropical region, e.g. maize, sorghum, etc.
- ❖ C₃ and C₄ pathway differ in the first product of carbon fixation. In C₃ plants, 3 carbon compound 3-phosphoglyceric acid (PGA) is produced, whereas, in C₄ plants, 4 carbon compound oxaloacetic acid (OAA) is produced.
- ❖ In C₄ plants also C₃ pathway is used in the formation of a glucose molecule
- ❖ C₄ plants have **Kranz anatomy** in leaves to tolerate high temperature. Large bundle sheath cells are present around vascular bundles of leaves
- ❖ Bundle sheath cells have thick walls, no intercellular spaces and have large chloroplasts
- ❖ Carbon fixation occurs in mesophyll cells. CO₂ acceptor is a 3 carbon compound phosphoenolpyruvate (PEP)
- ❖ The enzyme PEP carboxylase (PEPcase) catalyses the reaction. Mesophyll cell lack RuBisCO.
- ❖ The first product of carbon dioxide fixation is 4 carbon compound OAA
- ❖ OAA is then converted to other 4C acids like malic acid and aspartic acid. They are transported to bundle sheath cells. By decarboxylation in bundle sheath cells, CO₂ is released, which enters Calvin cycle.
- ❖ The 3-carbon acid is transported back to mesophyll cells
- ❖ The bundle sheath cells have RuBisCO but lack PEPcase.

Summary



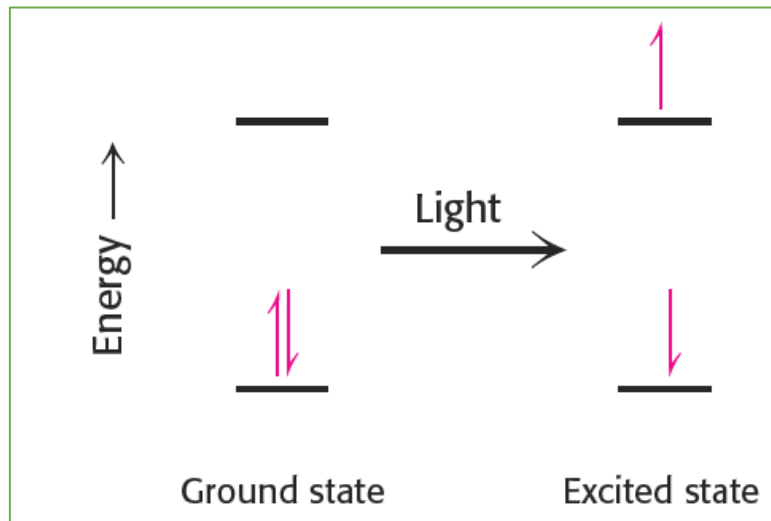
- ✓ The light reactions capture solar energy and use it to make ATP and transfer electrons from water to $NADP^+$.
- ✓ The Calvin cycle uses the ATP and NADPH to produce sugar from carbon dioxide.
- ✓ The energy that enters the chloroplasts as sunlight becomes stored as chemical energy in organic compounds.
- ✓ Sugar made in the chloroplasts supplies the entire plant with chemical energy and carbon skeletons for the synthesis of all the major organic molecules of cells.
 - ✓ About 50% of the organic material is consumed as fuel for cellular respiration in plant mitochondria. Some photosynthetic products are lost to photorespiration.

- Extras for understanding are given

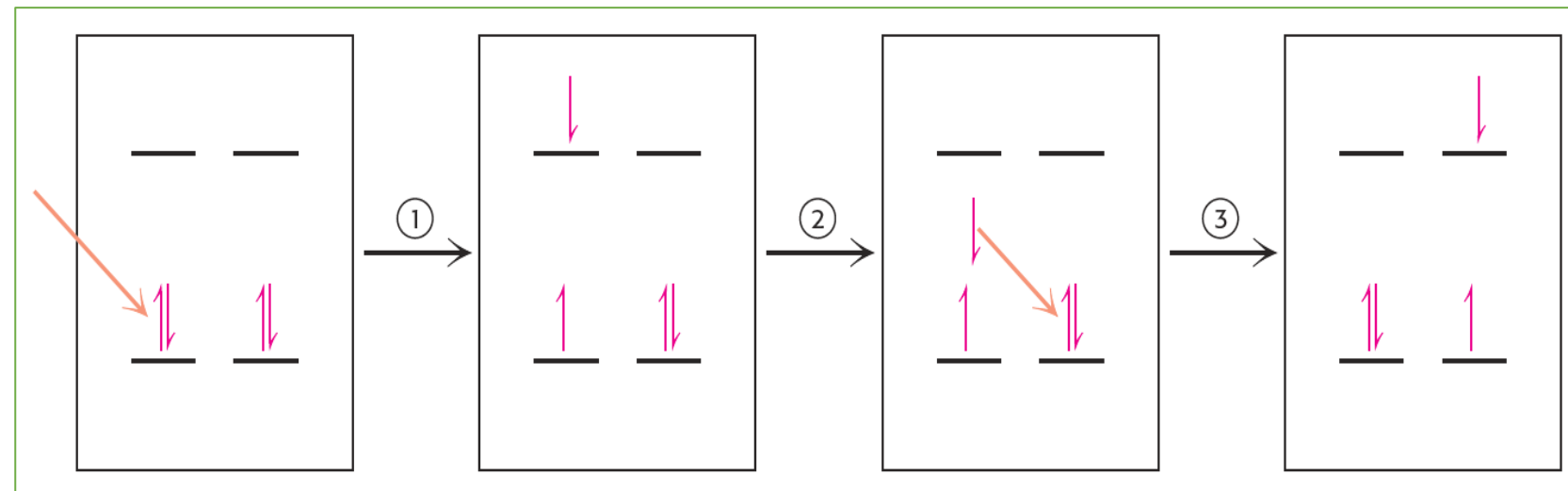
Light absorption



Light absorption



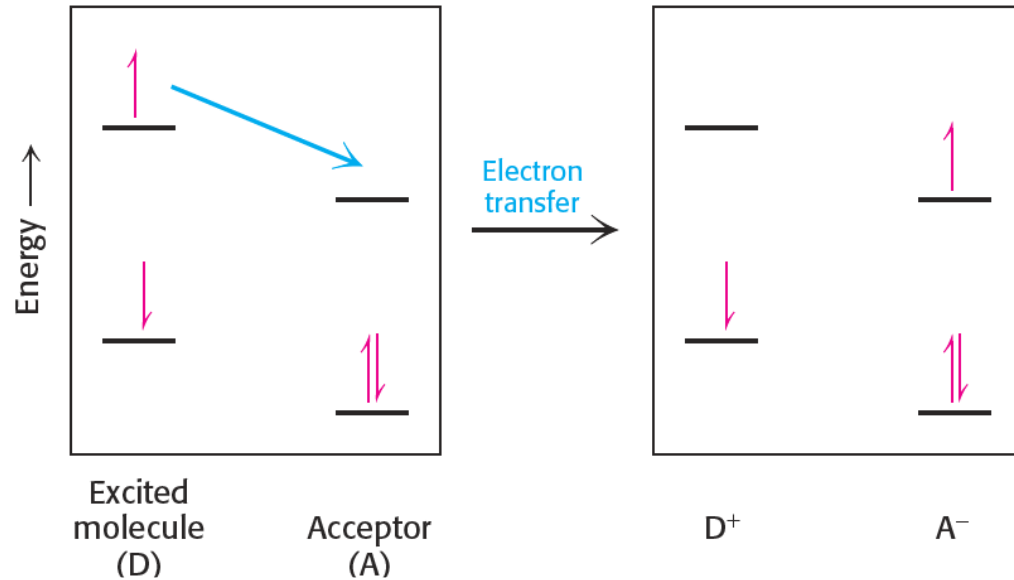
Resonance energy transfer



The absorption of light leads to the excitation of an electron from its ground state to a higher energy level. The horizontal lines represent different energy levels of the atoms in the pigment molecules.

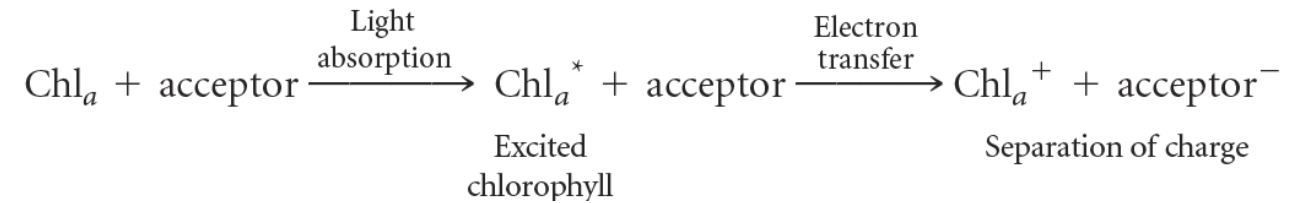
(1) An electron can accept energy from electromagnetic radiation of appropriate wavelength and jump to a higher energy state. (2) When the excited electron falls back to its lower energy state, the absorbed energy is released. (3) The released energy can be absorbed by an electron in a nearby molecule, and this electron jumps to a high energy state. The horizontal lines represent different energy levels of the atoms in the pigment molecules.

Photo-induced charge separation



If a suitable electron acceptor is nearby, an electron that has been moved to a high energy level by light absorption can move from the excited molecule to the acceptor.

In photosynthesis



Difference between PS-I and PS-II



PHOTOSYSTEM 1 VERSUS PHOTOSYSTEM 2

Photosystem 1 is located on the outer surface of the thylakoid membrane

Photosystem 2 is located on the inner surface of the thylakoid membrane

The photocenter is P700

The photocenter is P680

Pigments absorb longer wavelengths of light (>680 nm)

Pigments absorb shorter wavelengths of light (<680 nm)

Involved in both cyclic and non-cyclic photophosphorylation

Only involved in cyclic photophosphorylation

No photolysis of water occurs

Photolysis of water occurs

Main function is NADPH synthesis

Main function is ATP synthesis and hydrolysis of water

Contains chlorophyll B, chlorophyll A-670, Chlorophyll A-680, chlorophyll A-695, chlorophyll A-700 and carotenoids

Contains chlorophyll B, chlorophyll A-660, chlorophyll A-670, chlorophyll A-680, chlorophyll A-695, chlorophyll A-700, phycobilins and xanthophylls

Core is made up of psaA and psaB subunits

Core is made up of D1 and D2 subunits

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